

Technology
Arts Sciences
TH Köln

Cologne University of Applied Sciences

Faculty of Information, Media and Electrical Engineering

Bachelor's Thesis in Mediatechnology

Realtime Dereverberation Using Deep Learning

Echtzeit-Enthallung auf Basis von Deep-
Learning-Verfahren

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Start Date:	12.01.2026
Submission Date:	13.04.2026

We hereby confirm that this Bachelor's thesis is our own work and we have documented all sources and material used.

Cologne, 13.04.2026

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Transparency in the use of AI tools

AI tools, namely Microsoft Copilot, OpenAI ChatGPT 5.2 and 5.3-Codex as well as Claude Sonnet 4.6, were utilized in parts for code generation allowing for faster prototyping and analysis of data. These tools additionally provided writing support in the form of synonym suggestion and reformulation of passages. Machine translation of relevant content was performed using DeepL.

All AI-assisted outputs were reviewed and verified by the authors prior to inclusion.

Acknowledgements

Computations were performed with computing resources granted by RWTH Aachen University under project **thes2191**.

*“Well, I am going back into the open air, to
see what the wind and sky are doing!”*

— J.R.R. Tolkien, The Lord of the Rings

For Gondor!

Abstract

Reverberation is an inherent characteristic of real-world recording environments, and its removal (dereverberation) poses a significant challenge due to the highly non-linear, time-dispersive nature of room acoustics. While deep learning has enabled substantial progress in speech dereverberation, the generalization of these models to diverse audio signals is not well documented. In recent years metrics such as the SI-SNR and PESQ have been successfully utilized as loss functions. Their performance for dereverberation has not been systematically evaluated.

This thesis evaluates two state-of-the-art speech dereverberation models (Conv-TasNet and StoRM) for out-of-domain performance on diverse audio signals. Based on this evaluation, a novel dereverberation network was developed, built on the architecture of Conv-TasNet. Furthermore, the applicability of six fundamentally distinct loss functions for dereverberation is analyzed, concluding in the implementation of a novel scoring network.

Our results show a gap between speech and out-of-domain dereverberation performance in the investigated existing models. The evaluation of loss functions reports the SI-SNR as the strongest metric. A scoring network was successfully implemented and validated, showing strong predictive performance, outperforming the SI-SNR as a dereverberation indicator. While the SI-SNR was successfully used in the implementation of our dereverberation network, the scoring network failed to generate usable results.

Zusammenfassung

Nachhall ist ein typisches Merkmal realer Aufnahmeumgebungen, und seine Beseitigung (Enthallung) stellt aufgrund der stark nichtlinearen und zeitdispersiven Eigenschaften der Raumakustik eine erhebliche Herausforderung dar. Während Deep Learning erhebliche Fortschritte bei der Sprach-Enthallung ermöglicht hat, ist die Generalisierbarkeit dieser Modelle auf diverse Audioinhalte noch nicht gut dokumentiert. In den letzten Jahren wurden Metriken wie der SI-SNR und PESQ erfolgreich als Verlustfunktionen eingesetzt. Ihre Leistungsfähigkeit bei der Enthallung wurde jedoch noch nicht systematisch evaluiert.

Diese Arbeit evaluiert zwei aktuelle Modelle zur Sprach-Enthallung (Conv-TasNet und StoRM) hinsichtlich ihrer Leistung außerhalb ihres ursprünglichen Anwendungsbereichs bei vielfältigen Audiosignalen. Auf der Grundlage dieser Bewertung wurde ein neuartiges Enthallungsnetzwerk entwickelt, das auf der Architektur von Conv-TasNet aufbaut. Darüber hinaus wird die Anwendbarkeit von sechs grundlegend unterschiedlichen Verlustfunktionen für die Enthallung analysiert, was zur Implementierung eines neuartigen Bewertungsnetzwerks führt.

Unsere Ergebnisse zeigen eine Lücke zwischen der Sprach- und Out-of-Domain-Enthallungsleistung der untersuchten bestehenden Modelle. Die Bewertung der Verlustfunktionen ergab, dass SI-SNR die aussagekräftigste Metrik ist. Ein Bewertungsnetzwerk wurde erfolgreich implementiert und validiert und zeigte eine starke Vorhersageleistung, wobei es den SI-SNR als Indikator für die Enthallung übertraf. Während der SI-SNR erfolgreich bei der Implementierung unseres Enthallungsnetzwerks verwendet wurde, gelang es dem Bewertungsnetzwerk nicht, überzeugende Ergebnisse zu generieren.

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Glossary

Acoustics

AIR. Aachen Impulse Response	33
ASR. automatic speech recognition	1, 49
DCASE. Detection and Classification of Acoustic Scenes and Events	31
ISM. image source model	13, 14, 35, 36
RIR. room impulse response	13, 33, 35, 36, 78
STT. speech-to-text	1, 15, 31

Metrics

BSD. Bark Spectral Distortion	29
DI. distortion index	30, 37, 41, 76
KDE. kernel density estimation	40, 41, 42, 59, 80, 97, 98, 99
LCC. linear correlation coefficient	8, 9
LMS. least mean squares	22
MAE. mean absolute error	8, 27, 42, 59, 60, 66, 67, 69, 78, 81
MOS. mean opinion score	29, 42, 81
MSE. mean squared error	8, 2, 9, 27, 29, 38, 42, 46, 48, 52, 54, 55, 56, 59, 60, 66, 67, 69, 70, 76, 78, 79, 81
MSS. multi-scale spectral	46, 54, 55, 79
ODG. objective difference grade	30, 37, 41, 43, 47, 51, 59, 60, 66, 67, 69, 71, 76, 78
PAQM. Perceptual Audio Quality Measure	30

PEAQ. Perceptual Evaluation of Audio Quality	8, 30, 37, 39, 43, 44, 76, 78, 79
PESQ. Perceptual Evaluation of Speech Quality	5, 6, 8, 2, 9, 29, 38, 41, 43, 56, 67, 71, 76, 78, 79, 81
PSQM. Perceptual Speech Quality Measure	29
SDG. subjective difference grade	30
SI-SDR. scale-invariant signal-to-distortion ratio	28
SI-SNR. scale-invariant signal-to-noise ratio	5, 6, 8, 2, 28, 29, 38, 41, 42, 46, 52, 53, 54, 56, 61, 62, 66, 67, 69, 70, 71, 72, 75, 76, 77, 78, 79, 80
SRCC. Spearman's rank correlation coefficient	8, 9
ViSQOL. Virtual Speech Quality Objective Listener	39

Neural Networks

BLSTM. bidirectional long short-term memory	8, 17
CNN. convolutional neural network	7, 8, 9, 3, 16, 46, 59, 65, 66, 81
CV. computer vision	31
DAE. denoising autoencoder	19, 79
GRU. gated recurrent unit	17
LLM. large language model	31
LSTM. long short-term memory	17
RNN. recurrent neural network	7, 8, 16, 17, 18, 79
ReLU. rectified linear unit	23, 43, 47, 69
SGD. stochastic gradient descent	25
TCN. temporal convolutional network	7, 5, 16, 17, 18, 51

Signal Processing

ERB. equivalent rectangular bandwidth	7
STFT. short-time Fourier transform	3, 4, 5, 7, 47

1 Introduction

Reverberation is apparent in most audio signals, as it is an inherent characteristic of recording environments. It is caused by late reflections ($> 50 - 80$ ms) that overlap with the direct sound in the diffuse sound field [Kuh25]. Studies have shown that reverberation is an important auditory cue that informs the listener about environmental factors [TM16]. Depending on the application, reverberation can be an attractive addition to the auditory signal, such as in music [Nay14] or speech performances.

The inverse task, aptly named dereverberation, is the process of removing reverberant components from an audio signal recorded under reverberant conditions. Thus retaining only the direct sound of the recording. This was only made feasible due to recent advancements in the field of deep learning. Historically, this task was approached using standard signal processing techniques that involved coherence estimation, suppression rules based on thresholds and gain functions [ABB77, BC82].

In many modern applications, dereverberation is highly desirable. We divide use cases into two main categories: *offline* and *live* processing. Offline applications do not strictly require real-time operation, although real-time capability may still be beneficial.

1.1 Motivation

Studies have shown that the adverse effects of reverberation, mainly the temporal smearing of target speech and background noise, masking, coloration, and signal-to-noise degradation [CLG22, Fig+25, Kuh25] significantly degrade human as well as automatic speech recognition (ASR) or speech-to-text (STT) [Neu+10, Pug+21]. These effects also negatively affect the overall quality of diverse audio signals, such as in music remixes and film post-production, where excessive room reverberation can reduce audio clarity and limit creative flexibility. While the above-named offline applications are not in need of real-time processing, live applications, as they are used in interactive scenarios such as video conferencing,

speech recognition systems, and live music performance, impose strict constraints on processing latency and computational efficiency.

1.2 Problem

Artificial reverberation can be added to audio signals with comparatively simple signal processing techniques; the inverse task of removing or reducing existing reverberation is significantly more complex [Att+00]. Reverberation is a time-dispersive and highly non-linear process, where direct sound and multiple delayed reflections overlap in both time and frequency [Dat97]. This overlap makes a clear separation between the original (dry) signal and the reverberant components (wet) difficult and, for a long time, was considered practically unsolvable using classical digital signal processing methods [Bra98].

We address several open challenges in this thesis. First, it is unclear how well established dereverberation architectures generalize to diverse audio signals such as music and mixed content [LM18]. Current dereverberation models utilize different loss functions such as mean squared error (MSE), scale-invariant signal-to-noise ratio (SI-SNR) or Perceptual Evaluation of Speech Quality (PESQ) [Lem+23, LM19, Rad21] whose indication of dereverberation performance in diverse audio signals is not well documented. It is therefore unclear whether these loss functions are applicable for our use case or if other qualitative metrics would improve results. Second, there are time-domain and frequency-domain approaches, which can be investigated in terms of audio quality, computational complexity, and latency. Third, real-time applicability imposes strict latency limits (e.g., below 50 ms) [Sch+24] that strongly influence network architecture, window size, and sampling rate.

1.3 Objectives

The central objective of this thesis is to explore deep-learning-based dereverberation methods. As our primary question, we ask whether a model can be designed to operate in real-time while maintaining perceptually convincing audio quality for a wide range of audio signals, including music, speech, and other noises (e.g., vehicle

or animal sounds). Solutions for multiple performance constraints are surveyed, mainly the dataset, domain, and loss function.

The dataset processing and curation process is discussed regarding bias, variance, and sample quality.

Time-domain and frequency-domain neural network approaches are investigated by evaluating their qualitative performance and analyzing their suitability for low-latency, real-time applications through comparison of domain-specific literature and our own neural network implementations.

The performance of different quality metrics is assessed, and their qualitative performance for use as loss functions is examined, leading to our own implementation of a loss neural network.

1.4 Background

As mentioned in Section 1.2, dereverberation is a problem that has recently been addressed using machine learning methods. Three main approaches have emerged, each with their own strengths and shortcomings.

Classic signal processing implementations aim to enhance audio through subtraction of a predicted noisy signal [Bol79] or Wiener filtering [Mad+13]. Further developing these methods has introduced deep filtering approaches, utilizing machine learning to optimize frequency filter parameters [Zha+21]. Section 2.3 examines the “DeepFilterNet” model.

Other literature can be divided into time-domain or frequency-domain approaches, based on the representation of audio used. Time-domain methods operate directly on the audio waveform and use individual samples as input into the network. Frequency-domain approaches, on the other hand, transform the input waveform into a frequency-domain representation first, commonly using the short-time Fourier transform (STFT).

This has the advantage of being able to rely on convolutional neural network (CNN) architectures, which allow for faster processing, lower parameter counts, and preservation of local correlations in time and frequency that can be

captured efficiently by two-dimensional filters (see Section 3.2.1.1). Time-domain approaches allow for longer temporal contexts, allowing the model to better understand temporal dependencies (see Section 3.2.1.2 and Section 3.2.1.3).

The two models compared, as well as the one presented in this thesis, can be classified in the latter two categories. StoRM [Lem+23] is a frequency-domain model, making use of the STFT, while Conv-TasNet [LM19] and our novel implementation both operate in the time-domain.

Recent literature regarding dereverberation and adjacent problems (e.g., speaker–source separation and denoising) has focused on improving model architectures and loss functions for better performance. Notably, Fu et al. (2021) utilize a novel neural network-based loss for qualifying source separation performance introduced by Fu et al. (2018). This approach is further discussed in Section 2.4 and has motivated our own implementation of a perceptual quality loss network (see Section 4.3).

2 Related Work

2.1 Conv-TasNet

Leo Kling

Conv-TasNet extends the time-domain audio separation paradigm introduced by TasNet [LM18]. TasNet was proposed as an alternative to short-time Fourier transform (STFT) based source separation, motivated by the phase–magnitude decoupling of spectral methods, the fixed analysis representation, and the latency introduced by time-frequency decomposition. Instead of predicting masks on a spectrogram, TasNet learns an encoder–decoder representation directly on the waveform and performs separation in this latent space. Conv-TasNet retains this general idea but replaces the original recurrent separator with a fully convolutional architecture designed to model long temporal context more efficiently [LM19].

Architecturally, Conv-TasNet consists of a learned linear encoder, a masking network, and a linear decoder. The encoder transforms the input waveform into a latent representation, the separator estimates multiplicative masks for this representation, and the decoder reconstructs the target waveform from the masked features. The key architectural contribution of Conv-TasNet is the use of a temporal convolutional network (TCN) with stacked dilated one-dimensional convolution blocks as the separation module (see Section 3.2.1.3). This allows the model to capture long-range temporal dependencies while remaining compact and fully convolutional. In the original paper, the model is trained with SI-SNR and evaluated on single-channel, speaker-independent speech separation, where it outperforms preceding time-frequency masking approaches and even surpasses ideal time-frequency magnitude masking on the WSJ0 benchmark [LM19].

For this thesis, Conv-TasNet is relevant because it combines strong reported speech-domain performance with low model complexity and a comparatively small minimum latency, making it a plausible candidate for real-time capable dereverberation systems. At the same time, the scope of the original work remains limited to speech separation. The paper does not investigate dereverberation on diverse broadband material such as music, environmental sounds, or mixed acoustic

scenes. Therefore, Conv-TasNet here serves not as a solved answer to the research problem but as a strong real-time speech-domain baseline whose transferability to diverse-signal dereverberation must be evaluated separately.

2.2 StoRM

Leo Kling

StoRM is a diffusion-based stochastic regeneration model for speech enhancement and dereverberation [Lem+23]. In contrast to purely predictive enhancement systems, it follows a fully generative formulation in which the target signal is refined through a reverse diffusion process. The model combines a predictive estimate with stochastic generative refinement, aiming to retain the robustness of predictive methods while benefiting from the higher sample quality often associated with diffusion-based generation.

Architecturally, StoRM relies on score-based estimation during the reverse diffusion process and uses a predictive model as a guide for the generative reconstruction. In this sense, the score estimation component is conceptually related to the scoring used in this thesis, although it serves a different role than the perceptual loss network introduced later. The main motivation of the original paper is to reduce artifacts that may arise in purely generative diffusion systems while still producing very clean speech restoration results. At the same time, the paper explicitly notes the high computational burden of diffusion-based inference, since multiple reverse steps are required instead of a single forward pass.

For this thesis, StoRM is relevant because it demonstrates that a generative speech-domain approach can achieve strong dereverberation quality under its intended conditions. However, the scope of the original work remains limited to speech signals and a 16 kHz setting, which limits the representable spectrum to 8 kHz by the Nyquist theorem, which is substantially below the upper range of human hearing and therefore does not preserve the full audible bandwidth of music and other broadband audio material [ISO19, Pum50, Sha49]. In addition, the model is not designed around strict real-time constraints, making it less suitable as a direct answer to the low-latency objective of this thesis. StoRM therefore serves here not as a solved solution to real-time dereverberation of diverse audio, but

as a high-quality speech-domain reference whose transferability to broader signals and latency-constrained applications must be evaluated separately.

2.3 DeepFilterNet

Jonathan Kron

DeepFilterNet is a two-stage speech enhancement framework utilizing deep filtering [Sch+22]. Deep filtering is based on the idea that a neural network estimates a complex mask, which is applied to the STFT representation of a signal. With an appropriate loss function, this mask can be learned to perform source extractions or other tasks like dereverberation [MH20].

DeepFilterNet expands on that idea: “Instead of using a complex mask that is applied per TF-bin, [...] a combination of real-valued gains and a deep filter enhancement component [is used].” [Sch+22]. Given a noisy signal, DeepFilterNet first transforms it into the frequency-domain using the STFT. Sampling rates up to 48 kHz and window sizes between 5 ms and 30 ms are supported. Using the frequency-domain representation, equivalent rectangular bandwidth (ERB) features mimicking human perception are computed. Both the complex features of the STFT and the ERB features are used as inputs for the model. A mask is trained on the complex STFT values, and an encoder–decoder network is trained on the ERB features, which is later used to weight the complex mask. DeepFilterNet takes advantage of the fact that noise and speech signals usually have a smooth spectral envelope. This allows for a computationally cheap encoder–decoder network.

For this thesis, DeepFilterNet is relevant because it shows how low-complexity speech enhancement can be implemented in the frequency-domain. Regarding audio sample rate, the scope of this project even exceeds ours, but latency is still an issue as window sizes smaller than 5 ms are not supported. Furthermore, since DeepFilterNet’s architecture exploits the spectral properties of speech, adapting it for use with diverse audio content proves challenging.

2.4 Quality Net

Jonathan Kron

Most speech or audio assessment tools need a reference signal. This requirement “considerably restricts the practicality of such assessment tools” [Fu+18]. Fu et

al. (2018) introduce an end-to-end, non-intrusive speech quality evaluation model named quality net, that aims to predict PESQ scores without a reference signal.

Architecturally, Quality-Net is based on bidirectional long short-term memory (BLSTM), which is an improvement to standard recurrent neural networks (RNNs) (cf. Section 3.2.1.2), where sequential data is processed in both forward and backward directions using two separate hidden layers (cf. Figure 2.1).

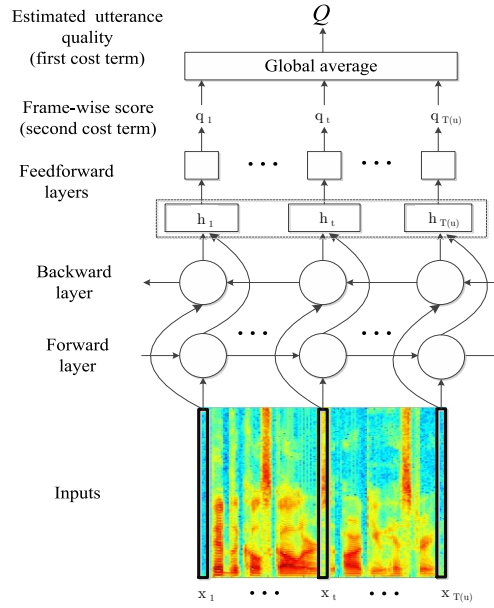


Figure 2.1: Proposed Quality-Net for end-to-end, non-intrusive speech quality assessment, taken from Fu et al. (2018).

Quality-Net was trained using 4620 utterances from the TIMIT [Gar+93] database. Utterances were divided into clean, noisy, and enhanced sets to learn different speech conditions. Noisy signals were generated by corrupting the original speech using 90 different noise types with signal-to-noise levels ranging from -10 dB to 25 dB. The enhanced utterances were first corrupted and then enhanced using a BLSTM based enhancement model.

Quality-Net’s performance was evaluated over 100 utterances using the MSE, linear correlation coefficient (LCC) and Spearman's rank correlation coefficient (SRCC) [Fu+18].

Table 2.1: Results of Quality-Net and the two-stage model replicated from Fu et al. (2018).

	MSE	LCC	SRCC	Variance of frame quality in clean speech
without constraint	0.1441	0.8559	0.8607	4.1128
with constraint	0.1266	0.8749	0.8807	0.2468

A constraint applies conditional weighting to each frame. This was done because noise can vary throughout an audio sample, and using a singular quality value for each frame of an audio sample could degrade model performance.

Table 2.1 shows good correlation but a considerably high MSE value, given that PESQ values lie between 0 and 4. These findings, among other things, motivated our own development of an audio assessment model, which is discussed in Section 4.3.

3 Theoretical Background

This section establishes the theoretical foundations necessary to understand the thesis. Acoustical fundamentals are covered first, addressing the origins of reverberation to ensure a clear understanding of the problem and to introduce the digital reverberators employed in the dataset creation. Sound quality is then described to convey the significance and practical application of the relevant metrics. The types of neural networks used throughout the thesis are subsequently introduced, followed by an overview of loss functions and their role in model training, highlighting the importance of selecting an appropriate loss function. Finally, the various quality metrics analyzed in Section 4.2 are presented.

3.1 Acoustics

3.1.1 Reverberation

Reverberation is the “hanging-on” of sound in a room after the exciting signal has been removed [Eve01]. It occurs naturally through late reflections of sound as described in Section 3.1.1.1 or artificially utilizing convolution (Section 3.1.1.3), delay networks (Section 3.1.1.2) or room simulations (Section 3.1.1.4) [Sch18, Val+12].

3.1.1.1 Natural Reverberation

Reverberation is caused by late reflections ($> 50 - 80$ ms) [Kuh25] of sound that overlap with the direct sound. This natural phenomenon can be observed in every environment that allows for sound reflections. An example is given by Everest (2001):

Figure 3.1 (A) shows a spherical sound source S and a listener L . Once S is energized, sound travels outward from S in all directions. Sound pressure at L instantly jumps to a value (D) that is less than that which left S due to spherical divergence and small losses in the air. Sound pressure continues to grow with each successive arrival of reflected components until an equilibrium is reached (cf. Figure 3.1 (B)).

Once S is turned off, the sound rays moving through the room lose their support, and with each successive reflection, they lose energy until they are considered dead. These reflections are also called the reverb tail. Figure 3.1 (C) shows the exponential decay of the first reflection components.

While the growth of sound is perceived as almost instant, the decay is slow.

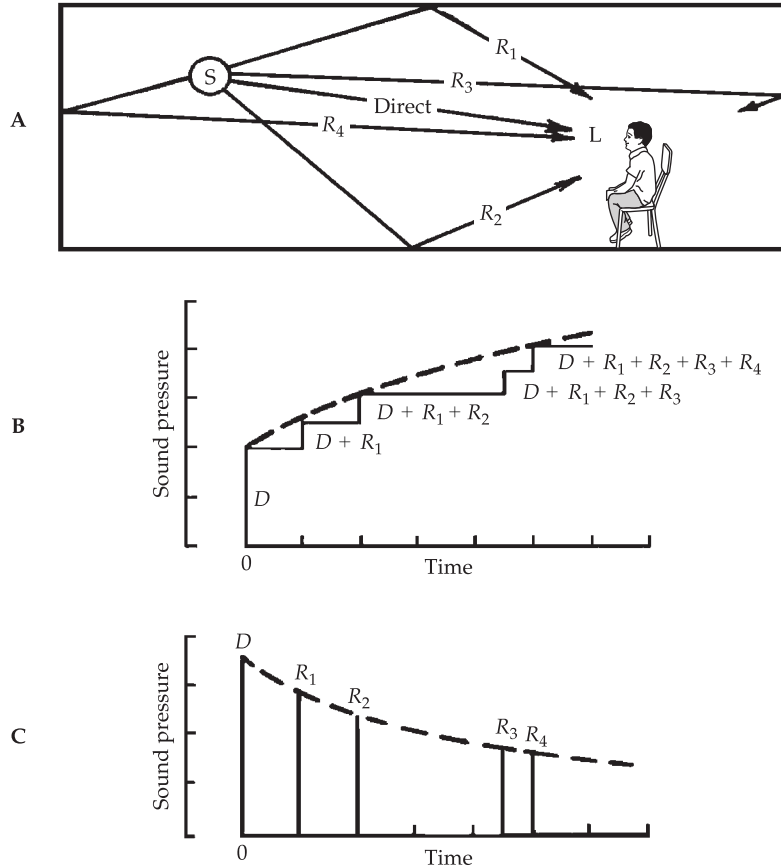


Figure 3.1: (A) Direct sound and exemplary first order reflections from source S arriving at listening position L . (B) The sound pressure at L grows stepwise. (C) The sound decays exponentially after the source ceases. Taken from Everest (2001).

Natural reverberation is characterized through the reverberation-time metric (RT). The RT_{60} indicates the time it takes for sound to decay by 60 dB after the source stops.

3.1.1.2 Delay Networks

In the 1960s, Schroeder first launched the field of digital artificial reverberation by introducing recursive comb filters and allpass filters as means for cheap simulation

of multiple echoes. The allpass filters achieve dense echoes with a flat amplitude response rate and have been used extensively in artificial reverberation [Moo79, RS97].

The Schroeder reverberator used in this thesis is called “Freeverb”, a public domain program by “Jezar at Dreampoint”. It found widespread adoption in the free-software world [Smi10].

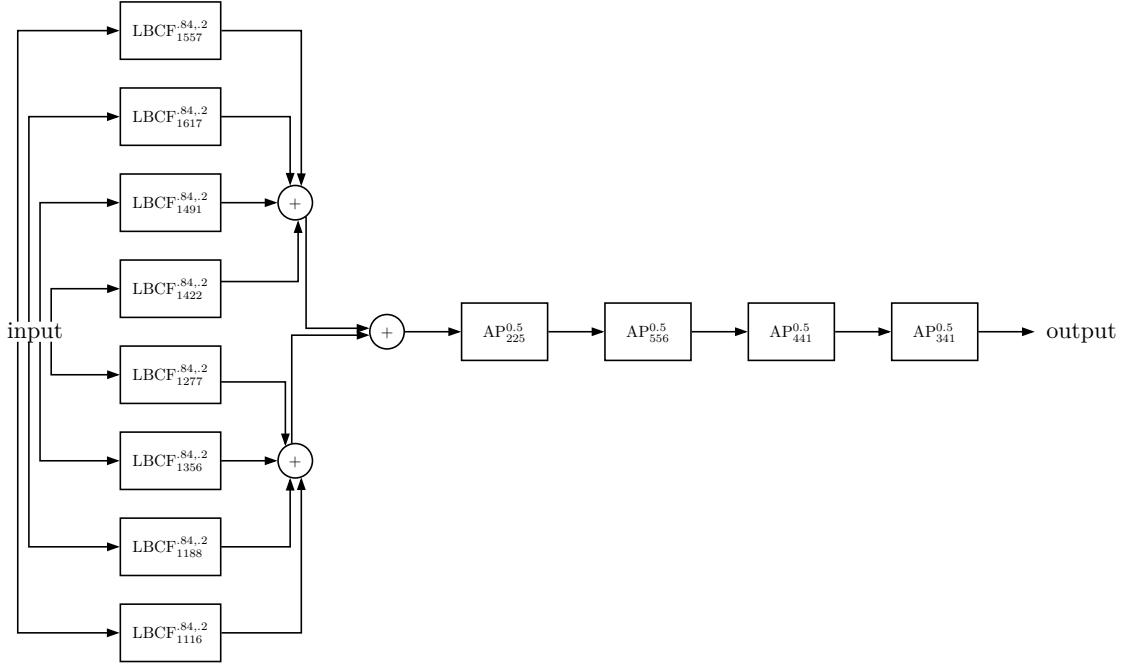


Figure 3.2: Freeverb block diagram (left stereo channel). Replicated from Smith (2010)

Figure 3.2 shows a block diagram of the left stereo channel of the Freeverb reverberator. The Schroeder-Moorer lowpass-feedback-comb-filters are denoted as $\text{LBCF}_N^{f,d}$. They are constructed using a delay line, an elementary functional unit that models acoustic propagation delay, whose output is low-pass filtered and summed with the delay line’s input [Smi10].

The transfer function of a lowpass-feedback-comb-filter is defined as

$$\text{LBCF}_N^{f,d} \triangleq \frac{z^{-N}}{1 - f \cdot \frac{1-d}{1-d \cdot z^{-1}} \cdot z^{-N}} \quad (3.1)$$

as introduced by Moorer (1979).

3.1.1.3 Convolutional Reverberation

Schroeder (1969) also proposed the idea of convolving a signal with a simulated room impulse response (RIR) to achieve natural-sounding reverberation. Hardware at that time could not calculate convolution of full RIRs and signals in reasonable times [Val+12]. Only in the late 1990s did hardware become powerful enough to support this approach to reverberation [Hug26]. Today, with advances in hard- and software, convolutions are considered a fast operation [MNT16, Sid20].

The convolution of two signals f and g is defined as [Win34]:

$$(f * g)(t) := \int_{-\infty}^{\infty} f(\tau)g(t - \tau) d\tau \quad (3.2)$$

Reverberation through convolution via RIRs is the most realistic way of generating synthetic reverb, as it mimics the scattering characteristics of a real-world room at the RIRs recording position [Far07].

3.1.1.4 Room Simulations

There are different approaches for modeling the acoustics of a room. The two main ones are either based on numerically solving the wave equation or on the assumptions of geometrical acoustics. Wave-based modeling techniques are able to provide the most accurate results. Their drawback is high computational cost. Therefore, faster but less accurate methods are often utilized [SS15].

Scheibler et al. (2018) introduce the pyroomacoustics library, which uses the image source model (ISM) to synthetically generate RIRs of specified positions in a room, which are then convolved (see Section 3.1.1.3) with the microphone signal to create the simulated output. The ISM method recursively mirrors sound sources across each plane of a room model, thereby creating image sources. The number of sources therefore grows exponentially. Each image source must be traced to the receiver position to assure a valid reflection path. It is then known, based on the distance of the image source, when the reflected sound should arrive at the listening position [AB79].

Once image source locations and visibilities have been determined, the impulse response is defined as

$$a_{\mathbf{r}}(\mathbf{s}_0, n) = \sum_{\mathbf{s} \in \mathcal{V}_{\mathbf{r}}(\mathbf{s}_0)} \frac{(1 - \alpha)^{\text{gen}(\mathbf{s})}}{4\pi \|\mathbf{r} - \mathbf{s}\|} \delta_{\text{LP}} \left(n - F_s \frac{\|\mathbf{r} - \mathbf{s}\|}{c} \right) \quad (3.3)$$

, where the microphone is placed at $\mathbf{r}$, the real source at $\mathbf{s}_0$ and the set of image sources is defined as $\mathcal{V}_{\mathbf{r}}(\mathbf{s}_0)$. F_s is the sampling rate, $\text{gen}(\mathbf{s})$ gives the reflection order of source $\mathbf{s}$, α is the wall absorption factor, c is the speed of sound, and δ_{LP} is the windowed sinc function [SBD18].

Another geometric modeling technique is ray tracing. It is based on the assumption that sound travels in straight lines and reflects off surfaces according to the law of reflection. Rays are emitted from the source and traced through the environment until they reach the listener or exceed a certain number of reflections [Vor08]. It is available in pyroomacoustics as a complement to the ISM.

3.1.2 Sound Quality

Section 3.3 describes different metrics that are used to qualify how well a corrupted audio signal sounds in comparison to its clean counterpart. Multiple of these metrics are perceptually motivated. This section gives an overview of parameters that impact the perception of audio. It is important to note that not all parameters listed can or should be used by quality metrics to qualify how good audio sounds (cf. Section 4.2).

There are many different signal corruptions that impact perceived sound quality. Some occur naturally, like reverberation or background noise, while others are of digital origin, like undersampling, insufficient bit depth, or packet loss in networks.

The impact of different signal corruptions differs between applications. Arguably the most important parameter in perceived speech quality is the amount of noise introduced in the signal, as speech intelligibility degrades significantly with lower speech-to-noise ratios [Lon06]. Speech intelligibility is also affected by excessive reverberation, as lower-level consonants are masked by the reverb tail. On the other hand, a completely dead room or an outdoor environment is not optimal for intelligibility, as the direct sound level may be too low to clearly hear

speech. Also affected by reverberation is music, as different genres benefit from different RT s. Absent reverberation makes music sound thin and weak [Eve01].

The human auditory complex perceives signals in the 20 to 20 kHz range [ISO19, MP04, Pum50]. A digital representation of a signal should therefore be sampled at a rate exceeding 40 kHz [Sha49]. While it is possible to represent speech using 8 kHz of bandwidth [ITU88] music and other broadband signals would lack important high-frequency informations.

While converting analog signals into digital representations, bit depth is another important factor. An 8-bit quantization allows for 49.93 dB of dynamic range. While speech requires a dynamic range of about 40 to 50 dB [DW40, Pav18], music sometimes features signal-to-noise ratios of up to 80 dB. Through the introduction of the compact disc, a quantization of 16 bits (offering 98.09 dB of dynamic range) has become the standard [Fre10].

Most of the above-named signal qualities also impact STT performance of neural networks [Neu+10, Pug+21]. A signal that is badly reverberated and noisy is harder to transcribe than an anechoic one. While humans have the added benefit of being able to deduce what was said by filling the gaps between understood words [Lon06] machines generally lack this ability.

Perceived sound quality is not only guided by signal corruptions but also by the overall frequency makeup of the signal. Although annoyance due to a sound can be highly subjective, it was shown that critical bands defined by the cochlea define frequency regions in which two simultaneously played sounds are perceived as dissonant [Lon06].

3.2 Neural Networks

Neural networks are parameterized function approximators composed of interconnected layers of simple computational units. During training, their weights and biases are iteratively adjusted so that the network maps an input to a desired output through minimizing a loss function [GBC16]. In practice, modern architectures differ mainly in how they organize these computations and which inductive

bias they impose on the data. For this thesis, the most relevant families are CNNs, RNNs, and temporal convolutional networks (TCNs).

3.2.1 Architectures

Leo Kling

3.2.1.1 CNN

CNNs are feed-forward neural networks that process structured inputs by applying learned convolution kernels over local neighborhoods. Instead of connecting every input element to every neuron, a convolutional layer reuses the same filter weights across the full input. This weight sharing reduces the number of parameters and makes the network particularly effective at detecting local patterns such as edges in images, harmonics in spectrograms, or short waveform structures [GBC16]. Stacking multiple convolution layers increases the receptive field, so deeper layers can combine simple local features into more abstract representations.

In audio machine learning, CNNs are widely used on time-frequency representations because spectrograms preserve local correlations in time and frequency that can be captured efficiently by two-dimensional filters. One-dimensional variants are also common when operating directly on waveforms or learned feature sequences. Their main advantages are computational efficiency, parameter sharing, and good parallelizability on modern hardware. A limitation is that long-range temporal dependencies are not represented explicitly and must instead be captured through depth, pooling, or large receptive fields [Gra+25, Kon+20].

3.2.1.2 RNN

RNNs are designed for sequential data. In addition to the current input, each recurrent step receives a hidden state that carries information from previous time steps, allowing the network to model temporal dependencies. This makes RNNs conceptually well suited for signals, text, and time series, where the interpretation of one sample often depends on earlier context, [GBC16, RHW86]. In audio applications, recurrent layers have been used for tasks such as speech enhancement, quality prediction, and sequence modeling because they can better aggregate information over longer temporal spans than shallow feed-forward models.

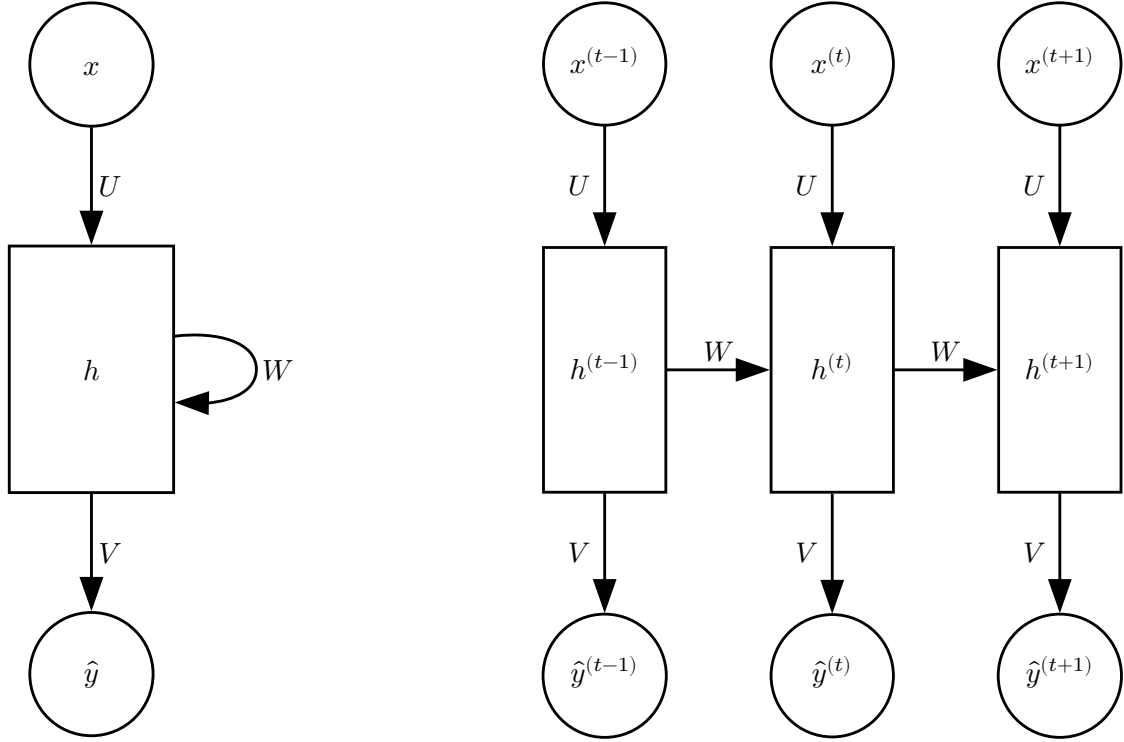


Figure 3.3: RNN architecture for an input sequence x , hidden connections parametrized by a weight matrix U and hidden-to-hidden recurrent connections parametrized by a weight matrix W . Shown on the right is the unrolled version of an RNN cell across three time steps. Replicated from Goodfellow et al. (2016).

Classical RNNs suffer from vanishing and exploding gradients when the dependency horizon becomes long. For this reason, gated variants such as long short-term memory (LSTM) [HS97], BLSTM and gated recurrent unit (GRU) networks [Cho+14] are commonly used in practice. These architectures regulate which information is stored, updated, or forgotten, improving training stability and long-term memory. Their main drawback is that recurrent processing is inherently sequential, which limits parallelization during training and inference compared to purely convolutional models [D  f+19, Fu+18].

3.2.1.3 TCN

TCNs adapt the convolutional idea to sequence modeling by applying one-dimensional convolutions along the temporal axis. To cover long contexts efficiently, they often use dilated convolutions, where filter taps are spaced apart by increasing

dilation factors (cf. Figure 3.4). This enlarges the receptive field without requiring very deep networks or large kernels. Depending on the application, TCNs can be implemented causally, where each output depends only on the present and past, or non-causally, where future context is also available [BKK18].

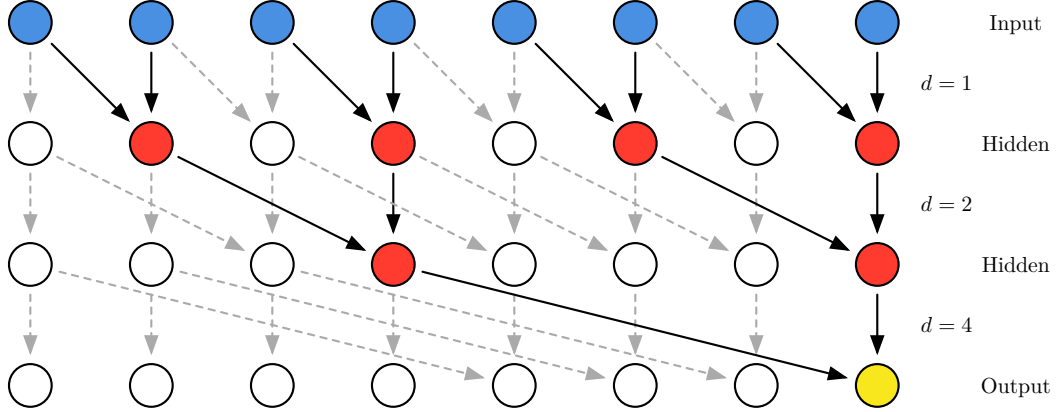


Figure 3.4: A dilated causal convolution with dilation factors $d = 1, 2, 4$. Replicated from Lee et al. (2021).

Compared to RNNs, TCNs retain the ability to model long temporal structure while remaining fully convolutional and therefore highly parallelizable. They also provide explicit control over receptive field size through kernel width, depth, and dilation schedule. This makes them attractive for audio tasks that require a compromise between temporal context and computational efficiency. In this thesis, TCNs are particularly relevant because Conv-TasNet uses a temporal convolutional network to estimate masks over an encoded waveform representation [LM19]. The basic principles of TCNs therefore form part of the architectural foundation for the dereverberation models discussed later.

3.2.1.4 Autoencoders

Autoencoders try to convert an input vector into a code vector of lower dimensionality and then reconstruct an approximation of the original input from this code vector [HZ93]. This network can be split into two parts, an encoder function

$$\mathbf{h} = f(\mathbf{x}) \quad (3.4)$$

and a decoder

$$\mathbf{r} = g(\mathbf{h}) \quad (3.5)$$

generating a reconstruction $\mathbf{r}$ of the original input $\mathbf{x}$ through a hidden layer $\mathbf{h}$ [LF87, GBC16].

Of particular interest for this thesis are denoising autoencoders (DAEs) which are a type of regularized autoencoder. Instead of minimizing some loss function

$$L(\mathbf{x}, g(f(\mathbf{x}))) \quad (3.6)$$

, where L is some loss function penalizing the difference between the input and the reconstruction, the denoising autoencoder minimizes

$$L(\mathbf{x}, g(f(\tilde{\mathbf{x}}))) \quad (3.7)$$

, where $\tilde{\mathbf{x}}$ is a stochastically corrupted version of the input. This forces the network to learn a more robust representation of the data, as it must be able to undo the corruption from a noisy version [GBC16]. In the context of dereverberation, DAEs can be used to learn a mapping from reverberant to clean speech by treating the reverberant signal as a corrupted version of the clean signal. The encoder learns to extract relevant features that are invariant to reverberation, while the decoder reconstructs the clean signal from these features.

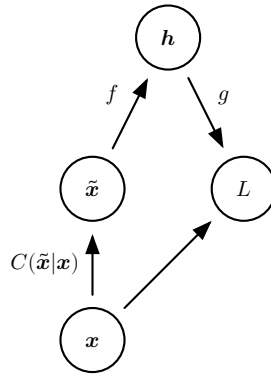


Figure 3.5: DAE training procedure. Replicated from Goodfellow et al. (2016).

Figure 3.5 shows the training procedure of a DAE. $C(\tilde{\mathbf{x}}|\mathbf{x})$ is the corruption process that gives a distribution over corrupted versions $\tilde{\mathbf{x}}$, given an original input $\mathbf{x}$. The

loss function L is then minimized with respect to the parameters of the encoder f and decoder g from training pairs $(\mathbf{x}, \tilde{\mathbf{x}})$ [GBC16].

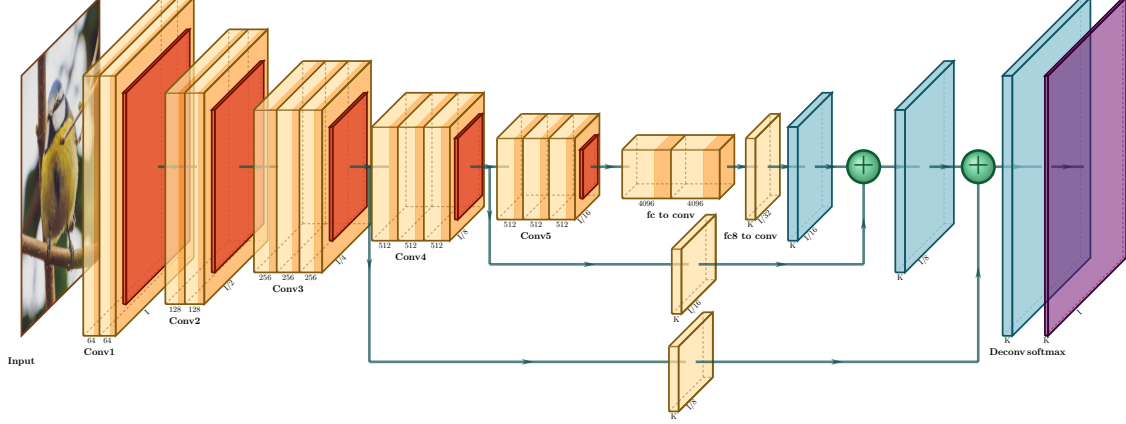


Figure 3.6: A U-Net architecture (FCN-8) [LSD15] with skip connections. The left side is the downsampling path where the input is processed through a series of convolutional and pooling layers, while the right side is the upsampling path where the feature maps are upsampled and combined with corresponding feature maps from the downsampling path through skip connections.

Autoencoders can also make use of skip connections between the encoder and decoder layers, first introduced by Srivastava et al. (2015), as shown in Figure 3.6. This allows the decoder to access high-resolution features from the encoder, which can prevent problems like vanishing gradients [He+15] and improve the landscape of the loss function [OP18], while keeping the depth and capacity of the network.

3.2.2 Supervised and Self-Supervised Learning

Leo Kling

Training data can be given to a network in different ways. The most common way is to use supervised learning, where each input data point is paired with a target output. The network learns to map inputs to their corresponding targets by minimizing a loss function that quantifies the error between the predicted output and the true target. We call this process **supervised learning** because one can imagine a supervisor who shows the network the correct answer for each input, and the network learns to mimic this behavior [GBC16].

Self-supervised learning is a form of unsupervised learning where, instead of relying on external labels, the data itself is used to generate supervisory signals.

This is often done by augmentation of the input data to formulate this signal. Self-supervised learning has been successful in recent years, especially in labeled data-scarce domains such as audio processing, as shown by Baevski et al. (2020). Two prominent paradigms within self-supervised learning are autoassociative and contrastive methods. In autoassociative learning, the network is trained to reconstruct its own input, forcing it to learn compact internal representations. Contrastive learning instead trains the network to pull similar samples together and push dissimilar ones apart in the representation space, without requiring explicit reconstruction [Bal+23].

3.2.3 Training of a Neural Network

Jonathan Kron

Section 3.2 states neural networks are iteratively trained through minimizing a loss function. The loss function is parameterized through an input-output function as well as the weights of the model. Optimizing the loss function means optimizing the weights.

We can imagine a multidimensional error landscape formed by the weights. To traverse this error landscape into a local minimum, we use the partial derivative of the loss function, also called the gradient. This process, coined gradient descent, is discussed in Section 3.2.3.3.

This gradient was historically computed analytically (see Section 3.2.3.1). Modern multi-million parameter networks make this approach impossible. To aid the process, modern networks use automatic differentiation based on the backpropagation method as introduced by Rumelhart et al. (1986).

3.2.3.1 Loss Function

A loss function, also called a cost function, is a qualitative function that is used to objectively measure model performance by calculating the deviation of the model's prediction from its ground truth counterpart. This deviation is mapped onto a real number that intuitively represents some error.

To introduce the application of loss functions, we want to discuss one of the earliest and simplest neural networks called Adaline [Wid60]. This single-layer neural network defines its input-output function as:

$$y(\mathbf{x}, \mathbf{w}) = \sum_{n=1}^N x_n w_n + b \quad (3.8)$$

where $\mathbf{x}$ is the input vector, $\mathbf{w}$ the weight vector, N the number of inputs, b some bias, and y the model output. Assuming that $x_0 = 1$ and $w_0 = b$ the output is simplified to:

$$y(\mathbf{x}, \mathbf{w}) = \sum_{n=1}^N x_n w_n \quad (3.9)$$

. Adaline uses the least mean squares (LMS) algorithm to define its loss function:

$$C(d, y) = (d - y(\mathbf{x}, \mathbf{w}))^2 \quad (3.10)$$

where d is the desired target. For analytical simplicity, the loss function is often denoted as:

$$\begin{aligned} C(d, y) &= \frac{1}{2} (y(\mathbf{x}, \mathbf{w}) - d)^2 \\ &= \frac{1}{2} (x_1 w_1 + x_2 w_2 + \dots + x_n w_n - d)^2 \end{aligned} \quad (3.11)$$

. The partial derivative can be calculated analytically (here for the first weight) by deriving the input-output function:

$$\begin{aligned} \frac{\partial C}{\partial w_1} &= \frac{1}{2} \cdot 2 \cdot (y(\mathbf{x}, \mathbf{w}) - d) \cdot y(\mathbf{x}, \mathbf{w})'_{w_1} \\ &= (y(\mathbf{x}, \mathbf{w}) - d) \cdot x_1 \end{aligned} \quad (3.12)$$

. The learning rule implementing this partial derivative is denoted as:

$$\mathbf{w} \leftarrow \mathbf{w} + \eta(d - y(\mathbf{x}, \mathbf{w}))\mathbf{x} \quad (3.13)$$

where η is some factor called the learning rate. This update rule implements gradient descent for linear regression. It should be noted that y is quadratic in the above loss function. Therefore no local minima are offered, and only a global minimum is approached [Ama93].

It can be concluded from the example above that analytical derivation of such loss functions becomes near impossible for complex input-output functions featuring non-linearities (activation functions) and millions of parameters. To solve this issue, the backpropagation algorithm is used.

3.2.3.2 Backpropagation and Autograd

Training a neural network happens in two steps. Initially the input is run through each of the network's functions. Through this process, called forward propagation, the neural network makes its best guess about the correct output.

Once an input-output pair is computed, the neural network calculates the gradient of the error function in regard to its guess by traversing backwards through its layers, collecting the derivatives of the error with respect to the parameters of the functions, which are later used to change each weight. This operation is also known as gradient descent (see Section 3.2.3.3).

The following section will discuss backpropagation as introduced by Rumelhart et al. (1986).

Expanding on the network example of Section 3.2.3.1 we define our multi-layer network as having a leftmost layer of input units, any number of intermediate layers, and a rightmost layer of output units. Connections within a layer or from right to left are forbidden, but connections can skip intermediate layers. The states of the units in each layer are determined by applying equations Equation (3.14) and Equation (3.15)

$$x_j = \sum_i y_i w_{ji} \quad (3.14)$$

$$y_j = \frac{1}{1 + e^{-x_j}} \quad (3.15)$$

, also called the forward pass, where Equation (3.15) is the sigmoid function, which today is often replaced by the rectified linear unit (ReLU) activation function.

The total error E is defined as (cf. Equation (3.11))

$$E = \frac{1}{2} \sum_c \sum_j (y_{j,c} - d_{j,c})^2 \quad (3.16)$$

where c is an index over all input-output pairs, j is an index over output units, y is the actual state of an output unit, and d is the desired state. The backward pass starts by computing the partial derivative of E with respect to x_j for each output unit. Differentiating Equation (3.16) for a single input-output pair

$$\begin{aligned} E &= \frac{1}{2} (y_j - d_j)^2 \\ &= \frac{1}{2} \left(\left(\frac{1}{1 + e^{-x_j}} \right) - d_j \right)^2 \end{aligned} \quad (3.17)$$

by applying the chain rule gives:

$$\begin{aligned} \frac{\partial E}{\partial x_j} &= \frac{\partial E}{\partial y_j} \cdot \frac{dy_j}{dx_j} \\ &= \frac{\partial E}{\partial y_j} \cdot y_j(1 - y_j) \end{aligned} \quad (3.18)$$

. This shows the affecting change is just a linear function of the states of the layer before, making it “easy” [RHW86] to compute how the error will be affected by changing states in the intermediate layers. For a weight w_{ji} the derivative is

$$\frac{\partial E}{\partial w_{ji}} = \frac{\partial E}{\partial x_j} \cdot y_i \quad (3.19)$$

. The output of the i^{th} unit, taking into account all emerging connections, results in

$$\frac{\partial E}{\partial y_i} = \sum_j \frac{\partial E}{\partial x_j} \cdot w_{ji} \quad (3.20)$$

. This shows how $\frac{\partial E}{\partial y}$ of the output layer can be computed when $\frac{\partial E}{\partial y_i}$ of the layer before is given. This procedure can therefore be repeated for each layer going backwards.

Historically these computations have been done manually by the researchers [Bay+15]. This task is tedious and error-prone. Here automatic differentiation algorithms are of assistance. PyTorch’s autograd system records all functional

computations involved in generating the network’s output in a directed acyclic graph, with input tensors as leaves and output tensors as roots. By traversing this graph from roots to leaves, gradients can be computed automatically via the chain rule [Aut26]. It is important to note that this automatic process requires every function to respect the input data’s need for a gradient. During computation, gradient calculation can be wrongfully disabled. This problem can occur when using another neural network as the loss function. This is further discussed in Section 5.4.

3.2.3.3 Gradient Descent

In Section 3.2.3.2, it is discussed how partial derivatives of the error function E can be calculated either manually or through the use of an automatic differentiation system.

Once ∇E is calculated, each weight can be adjusted so that the loss is further minimized (cf. Equation (3.13)). Through this process, called gradient descent, a local minimum is searched. Finding a global minimum is not necessary, as experts “[...] suspect that, for sufficiently large neural networks, most local minima have a low cost function value, and that it is not important to find a true global minimum [...]” [GBC16].

Rumelhart et al. (1986) introduce the simplest version of gradient descent as the accumulation of all gradients over all training examples and changing each weight by an amount proportional to the accumulated $\frac{\partial E}{\partial w}$. There are, in fact, improvements to this approach in the stochastic gradient descent (SGD) method, which approximates the gradient of the entire dataset over a small subset of training examples, also called minibatches. This lowers the computational cost of calculating a gradient over the entire dataset, which is especially useful when dealing with large amounts of data. It is not guaranteed that the SGD method arrives at a local minimum in a reasonable amount of time, but often a useful “low enough” loss is found [GBC16].

SGD is used during training of our perceptual quality network as well as the dereverberation network.

3.2.3.4 Taxonomy of Loss Functions

Section 3.2.3.1 and Section 3.2.3.2 made clear what impact a loss function can have on the training process of a neural network. Over the recent years, many different loss functions for different problem sets have been envisioned, each best suited for a specific input-output function with specific input-output data pairs [Cia+24].

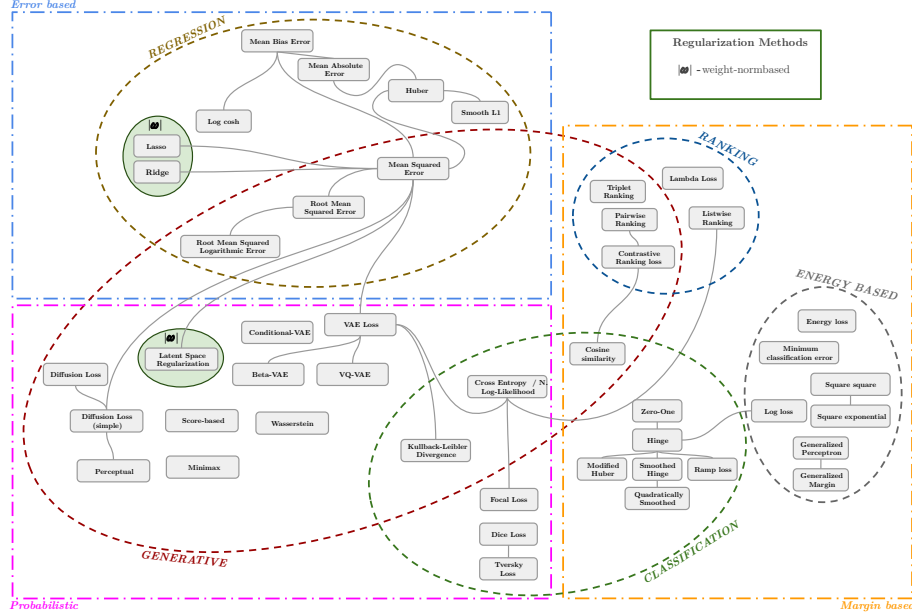


Figure 3.7: A taxonomy of loss functions taken from Ciampiconi et al. (2024).

Figure 3.7 shows a map that identifies five major tasks for which loss functions can be designed, namely regression, classification, and ranking, as well as generative and energy-based models. Optimization strategies for each task category are proposed, including error-based, probabilistic, and margin-based loss functions.

The problem of dereverberation can be attributed to a regressive task. It is shown that for the problem of regression, error-based loss functions are applicable. The following section will discuss error-based metrics that can be utilized as loss functions. An analysis of their performance as such is discussed in Section 4.2.

3.3 Quality Metrics

Jonathan Kron

The following section will present different quality metrics designed for comparative analysis of two input vectors. Going forward, the input vectors will be

considered signals as we are examining these measures from a signal processing standpoint.

$\mathbf{s}$ is defined as the ground truth, also named reference or true signal.

$\hat{\mathbf{s}}$ is defined as the predicted, also named test or processed, signal.

All subsequent measures are investigated for general usability in audio-adjacent machine learning tasks. Most are used in Chapter 6 for comparative evaluation of different neural networks. A discussion of usability as a loss function for a dereverberation neural network is found in Section 4.3.

3.3.1 MAE and MSE

The mean absolute error (MAE)

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n (s_i - \hat{s}_i) \quad (3.21)$$

measures the average absolute error between two signals. The mean squared error

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (s_i - \hat{s}_i)^2 \quad (3.22)$$

measures the average squared difference between the predicted and the ground truth signal. Although both the MSE and MAE were used successfully as loss functions in, for example, music source separation approaches [Déf+19, SED18, TM20] they fall short in generative and human-ear-centered tasks as both unfairly penalize shifts in time and amplitude of the predicted signal and do not conform to the equal-loudness levels as perceived by the human ear [Aco23] and therefore overweight the importance of low frequencies.

3.3.2 Correlation

The Pearson's product-moment coefficient is defined as:

$$\rho_{s,\hat{s}} = \text{corr}(s, \hat{s}) = \frac{\text{cov}(s, \hat{s})}{\sigma_s \sigma_{\hat{s}}} = \frac{\text{E}[(s - \mu_s)(\hat{s} - \mu_{\hat{s}})]}{\sigma_s \sigma_{\hat{s}}}, \text{ if } \sigma_s \sigma_{\hat{s}} > 0 \quad (3.23)$$

where σ_s and $\sigma_{\hat{s}}$ are the standard deviations, μ_s and $\mu_{\hat{s}}$ the expected values, and E the expected values operator [Ben+09]. The result of the Pearson coefficient can be interpreted as seen in Table 3.1, where negative values mean inverse association:

Table 3.1: Interpretation of the Pearson coefficient

$\rho_{s,\hat{s}}$	$\rho_{s,\hat{s}}$	Association Between Variables
+0.8 to +1.0	−0.8 to −1.0	Very strong association
+0.6 to +0.8	−0.6 to −0.8	Strong association
+0.4 to +0.6	−0.4 to −0.6	Moderate association
+0.2 to +0.4	−0.2 to −0.4	Weak association
+0.0 to +0.2	−0.0 to −0.2	Very weak or no association

The problem is that both input signals are assumed to be two random variables, which is technically not the case. Although correlation has been used successfully in computational audio tasks such as simultaneous sound event localization [Cor+19], using a statistical relationship to compare a reference to a test signal proved challenging (see Section 4.2).

3.3.3 SI-SNR

The scale-invariant signal-to-noise ratio first introduced as the scale-invariant signal-to-distortion ratio (SI-SDR) in [Rou+18] is defined as:

$$\text{SI-SNR} = 10 \log_{10} \left(\frac{\|as\|^2}{\|as - \hat{s}\|^2} \right), \text{ where } a = \frac{\hat{s}^T s}{\|s\|^2} \quad (3.24)$$

. The SI-SNR and SI-SDR have identical formulas but different interpretations of the corruption applied to the signal $\hat{s}$ (noise or distortion). The SI-SNR measures the level of noise in the predicted signal in a way that is invariant to amplitude scaling of the signals. It has been used successfully in dereverberation tasks [LM19] but while providing invariance to signal scaling, it too does not conform to the perceived loudness of the human ear nor provide invariance to signal shifting.

It is also worth noting that several related variants exist, including the Source-to-Artifact Ratio (SAR), Source-to-Interference Ratio (SIR), Source-to-Distortion Ratio (SDR), and Signal-to-Noise Ratio (SNR), each available in Scale-Invariant (SI) forms. These metrics are widely used in source separation and speech enhancement, and are all rooted in the conventional definition of the SNR [VGF06].

3.3.4 PESQ

Answering the shortcoming of metrics like the MSE and SI-SNR, the Perceptual Evaluation of Speech Quality (PESQ) model (a successor to the Bark Spectral Distortion (BSD) and Perceptual Speech Quality Measure (PSQM) models) is both invariant to signal scaling and shifting. In regard to its predecessors, PESQ produces more accurate scores when the signal is affected by coding distortion, packet loss, background noise, filtering, or variable delay. The PESQ score reflects speech quality on a continuous scale ranging from 1 to 5 (cf. Table 3.2).

Table 3.2: The Absolute Category Rating scale used by mean opinion score (MOS)/PESQ

Rating	Label
5	Excellent
4	Good
3	Fair
2	Poor
1	Bad

The scale shown in Table 3.2 corresponds to the MOS scale. During analysis, the signal is mapped into a representation of perceived loudness in time and frequency through a psychoacoustic model based on the bark scale [Rix+01], which is a psychoacoustical scale on which equal distances correspond with perceptually equal distances [Zwi61], therefore assuring conformity with the human auditory system (cf. Figure 3.8).

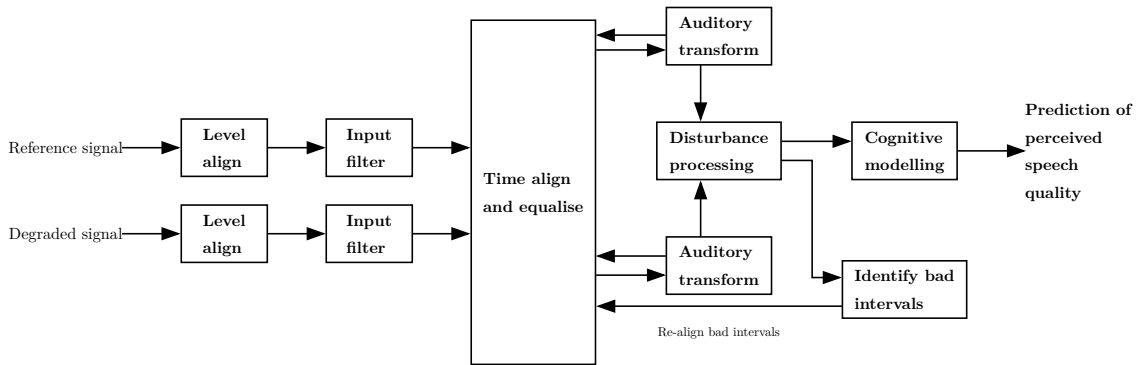


Figure 3.8: Structure of Perceptual Evaluation of Speech Quality (PESQ) model taken from Rix et al. (2001).

3.3.5 PEAQ

The Perceptual Evaluation of Audio Quality (PEAQ) model is based on the Perceptual Audio Quality Measure (PAQM) model and has been an ITU-R recommendation since 1999 [Rix+01]. In general, it compares two time-aligned signals, one processed and one original. Concurrent frames of each signal are transformed to a basilar membrane representation whose differences are further analyzed by a cognitive model [Thi+00] (cf. Figure 3.9). The two offered metrics, namely the objective difference grade (ODG) and distortion index (DI), are therefore not invariant to signal shifting, but they conform to the human perception of sound loudness. The objective difference grade (ODG) corresponds with the subjective difference grade (SDG) and indicates the audio quality of the tested signal on a continuous scale from -4 (very annoying impairment) to 0 (imperceptible impairment). The distortion index (DI) is a quality indicator like the ODG, except for its higher sensitivity towards very low signal qualities [KAD17].

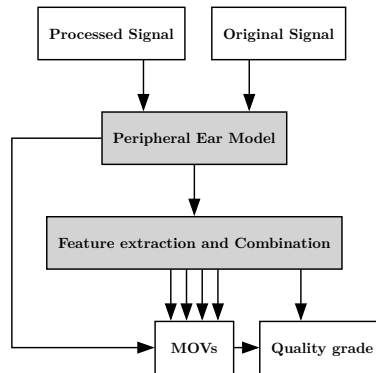


Figure 3.9: High-level representation of the Perceptual Evaluation of Audio Quality (PEAQ) model taken from Thiede et al. (2000).

4 Methodology

This chapter outlines the methodology underlying the thesis. It begins with the rationale behind the collection and creation of the dataset, followed by an analysis of applicable loss functions. Finally, the design and implementation of the three proposed networks are described: the perceptual quality network, the objective quality network, and the dereverberation network.

4.1 Dataset

Other machine learning fields, mainly computer vision (CV) and large language models (LLMs), have long been trained on publicly available diverse datasets [Den+09], namely mC4, MassiveText, or the Wikipedia dataset [Nav+25].

As shown in Chapter 2, previous work in the field of audio dereverberation has generally focused on speech signals. As this limitation is the same for many audio-based machine learning problems (e.g., multi-speaker separation, noise cancellation, and STT), many of the most used large audio datasets, like LibriSpeech or WSJ0, consist only of speech signals that are reduced in bandwidth as well as language diversity and recorded in anechoic conditions [Gar+07, Pan+15, Ric+24].

Datasets of diverse audio signals have emerged from audio classification problems. Early examples are private, self-collected datasets of individual researchers [Ell01, Woo92]. Over the recent years, interest in audio classification has surged, as can be seen in the amount of entries in the Detection and Classification of Acoustic Scenes and Events (DCASE) challenge series that increased from 31 in 2013 to 428 in 2023 [Mes+24]. The DCASE has also been a major influence in the increase of publicly available datasets, as prior to the DCASE challenges, only a limited amount were available, most notably RWCP [Smi10].

The current largest dataset of diverse audio signals is Google’s fittingly named AudioSet, containing over 5,800 hours of audio recordings with 527 classes of annotated sounds [Gem+17]. These recordings are 10-second clips drawn from YouTube videos. Building on top of the AudioSet classes, the FSD50K dataset

contains 100 hours of audio composed of 51,197 individual samples [Fon+22] taken from the “freesound.org” audio sharing site. The FSD50K dataset is publicly available, while AudioSet released embedding features of the raw audio data, necessitating a private download from YouTube. Both datasets are human-labeled, while AudioSet specifies that sounds are human-verified and classes are suggested using YouTube metadata.

4.1.1 Data Collection

Our proposed approach requires a diverse dataset of dry audio data. In total, 108,775 individual audio samples were collected, resulting in the following dataset:

Table 4.1: Dataset composition

Dataset	Number of Files	Length of Files
<i>AudioSet</i>	10790 (9.92 %)	44h 52m 34s (13.8 %)
<i>FSD50K</i>	46753 (42.98 %)	107h 34m 25s (33.1 %)
<i>LibriSpeech</i>	51232 (47.1 %)	172h 17m 49s (53.1 %)
<i>Total</i>	108775	324h 44m 49s

Diverse audio data from the AudioSet and FSD50K datasets were downloaded in 44.1 kHz. Both datasets were used to eliminate any bias occurring in one of the datasets (e.g., YouTube compression artifacts). The LibriSpeech dataset [Pan+15] includes English utterances recorded in anechoic conditions and sampled at 16 kHz. These were included in hopes of giving speech signals a greater weight, as we felt clean speech was underrepresented in the other datasets.

A final dataset split of 70% training, 15% validation, and 15% testing data was decided. Each sample was randomly assigned to one subset, allowing for equal distribution of the entire dataset (cf. Table 4.1) in each subset.

To assure reproducibility, the file name of each sample is hashed using MD5 [Riv92]:

$$\begin{aligned}
 h &= \text{MD5}(\textit{name}) \\
 r &= \frac{\text{int}(h_{0\dots3})_{\text{LE}}}{2^{32} - 1}
 \end{aligned}
 \tag{4.1}$$

The first 4 bytes are then used to assign the split:

$$\text{split}(r) = \begin{cases} \text{train} & \text{if } r < S_{\text{train}} \\ \text{val} & \text{if } S_{\text{train}} \leq r < S_{\text{train}} + S_{\text{val}} \\ \text{test} & \text{otherwise} \end{cases} \quad (4.2)$$

With $S_{\text{train}} = 0.7$ and $S_{\text{val}} = 0.15$. A final distribution, as can be seen in Table 4.2, was achieved.

Table 4.2: Subset composition

Subset	Number of Files	Length of Files
<i>Train</i>	76234 (70.08 %)	227h 45m 06s (70.13 %)
<i>Validation</i>	16120 (14.82 %)	48h 16m 47s (14.87 %)
<i>Testing</i>	16421 (15.1 %)	48h 42m 54s (15.0 %)
<i>Total</i>	108775	324h 44m 49s

Another dataset of RIRs was gathered, which was later in part used for reverberation purposes. The RIRs were collected from the Aachen Impulse Response (AIR) dataset [JSV09] as well as from the “Hybrid Reverb” plugin in Ableton Live 12¹. The AIR dataset contains 433 individual RIRs with different acoustical properties, such as reverberation time and room volume.

4.1.2 Dataset Creation

A supervised training approach (as explained in Section 3.2.2) was chosen to train both our dereverberation (cf. Figure 4.1) and perceptual loss model (cf. Figure 4.2). Labeling was done automatically through synthetic reverberation of the dry audio samples included in the dataset described in Section 4.1.1.

¹<https://www.ableton.com/en/packs/hybrid-reverb/>

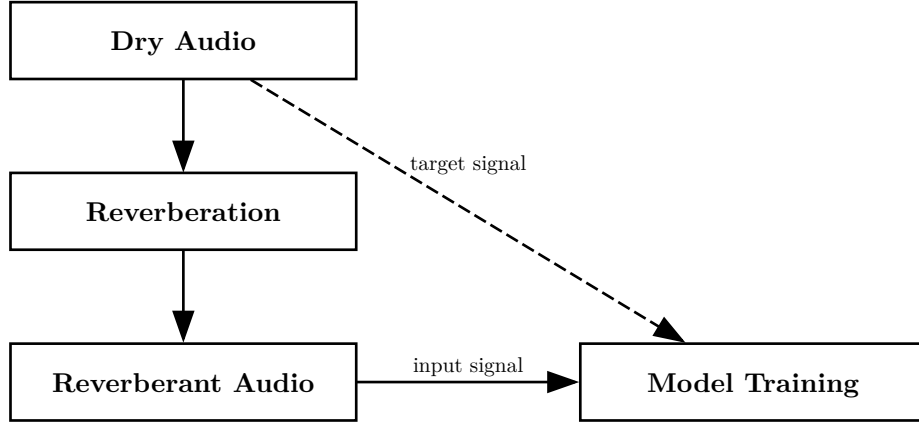


Figure 4.1: Dereverberation preprocessing pipeline

Additional labels used for the perceptual loss model (see Section 4.3) were saved during parameter-based reverberation (see Section 4.1.2.1) and calculated from the dry-reverberant-sample pairs (see Section 4.1.2.2).

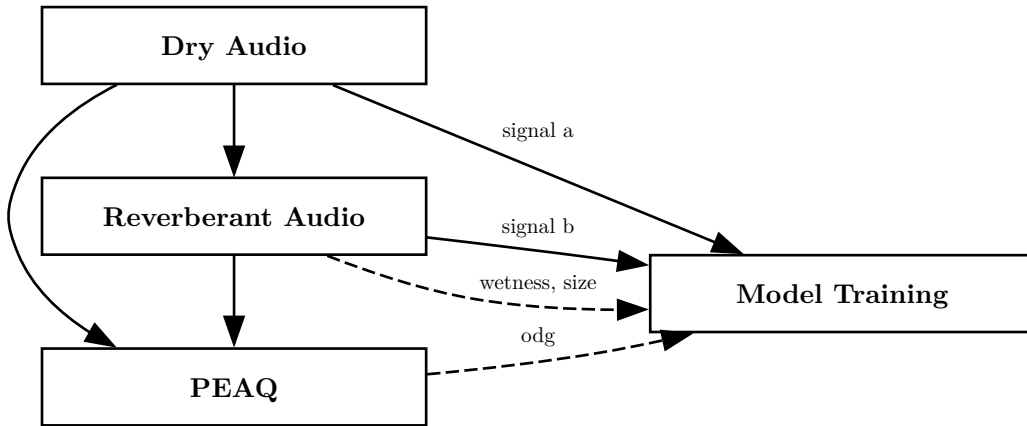


Figure 4.2: Perceptual loss preprocessing pipeline

This synthetic labeling approach is similar in concept to self-supervised training, where a supervisory signal is generated through augmentation (see Section 3.2.2). In the domain of computational audio, self-supervised approaches have shown great efficiency [Bae+20]. However, as our objective is neither autoassociative nor contrastive but a supervised regression from reverberant to dry audio, it cannot be classified as such (see Section 3.2.2).

4.1.2.1 Reverberation

To provide the model with reverberant audio signals, two kinds of preprocessing approaches were considered. The signals could either be reverberated “*live*”, after loading a sample into memory during training, or “*offline*” beforehand, saving compute time but sacrificing disk space.

The three primary approaches to digital reverberation [Sch18] are convolutional methods, delay networks, and computational acoustics (see Section 3.1.1).

Reverberation through convolution via RIRs is the most realistic way of generating synthetic reverb, as it mimics the scattering characteristics of a real-world room at the RIRs recording position [Far07]. Generally this comes at a higher computational cost and latency [MNT16, Sid20]. Unfortunately, convolution reverbs do not expose many parameters or controls, making labeling of samples that are fed to our loss network (see Section 4.3) difficult.

Parameter-based reverberation, like delay networks, is fast and requires little memory, but careful tuning is necessary to find configurations that sound realistic [Sch18, Sid20]. This gives us easy access to, e.g., size and wetness controls that we can use for labeling (see Section 4.3).

In computational acoustics, room simulations are used for reverberating audio [Lem+23]. Game engines such as Unity [Man+25] or libraries like pyroomacoustics [SBD18] can be used to simulate rooms with different sizes, materials, and microphone placements. This is done either by trying to solve the wave equation by the discretization of the space, geometric solutions like the ISM [AB79], or ray tracing [Vor08]. While this is attempting to recreate an acoustic space as close as possible, it is also the most computationally expensive and not possible to do live or offline for our amount of data. Unity’s processing is also done in real time, which makes it not feasible, as the runtime would be about 324 hours (cf. Table 4.1). In computational acoustics, room simulations are used for reverberating audio [Lem+23]. Game engines such as Unity [Man+25] or libraries like pyroomacoustics [SBD18] can be used to simulate rooms with different sizes, materials, and microphone placements. This is done either by trying to solve the wave equation by the

discretization of the space, geometric solutions like the ISM [AB79], or ray tracing [Vor08]. While this is attempting to recreate an acoustic space as close as possible, it is also the most computationally expensive and not possible to do live or offline for our amount of data. Unity’s processing is also done in real time, which makes it not feasible, as the runtime would be about 324 hours (cf. Table 4.1).

A first implementation was done using live processing in memory. All three approaches described above were implemented for an interchangeable framework. Using this, the original Conv-TasNet data loader was adjusted to use the RIR implementation.

After getting access to the RWTH Aachen CLAIX compute cluster, we pivoted to an offline dataset as compute time was of more importance than disk space. As mentioned, parameter-based reverberation was suited best for our own networks due to the access to size and wetness controls.

Specifically, we first used Valhalla Supermassive in the VST3 format, which was later abandoned, as it lacked Linux compatibility. We then chose a FreeVerb implementation [Smi10] in the `pedalboard` Python package [Sob23].

When processing the data from the dataset, the files were reverberated at their native sample rate and later upsampled to the sample rate used for training. The three sub-datasets have the following sample rates: AudioSet at 44.1 kHz, LibriSpeech at 16 kHz, and FreeSound at 44.1 kHz.

While 44.1 kHz is a fairly standard sample rate for consumer audio content [Pu06], 16 kHz only allows for an upper frequency of

$$f_{\max} = \frac{f_s}{2} = \frac{16 \text{ kHz}}{2} = 8 \text{ kHz} \quad (4.3)$$

to be represented, as shown by the Nyquist theorem [Sha49]. While this is technically enough to represent speech signals, which only need a bandwidth of 300 Hz to 3400 Hz [ITU88], the reverberation applied at the native sample rate, introduces bandwidth limitations in the reverb signal.

Section 7.1 discusses the distribution of reverberation times in the dataset.

4.1.2.2 Calculation of PEAQ Scores

For every dry-reverberant-sample pair, the PEAQ scores ODG and DI (see Section 3.3.5) were calculated. As the GStreamer implementation “GstPEAQ” was used [HZ15], GStreamer Python bindings were utilized to automate this process [GSt26]. This approach meant we needed both reference and test files written to disk, making a live implementation not feasible. All samples were upsampled to 48 kHz for use with PEAQ.

4.1.2.3 Non-Silent Parts

Using the Python package Pydub [Rob26] non-silent ranges of all samples of the training subset (see Section 4.1.1) were analyzed. A sample is considered silent if its level is below -40 dBFS. A range of samples is considered silent once it is longer than 100 samples.

Using this formula $d_{\text{full}} \approx 227.75$ hours of training data was examined. The duration of all non-silent ranges was $d_{\text{non_silent}} \approx 149.77$ hours leaving $d_{\text{silent}} \approx 77.98$ hours of silence or about

$$\frac{d_{\text{silent}}}{d_{\text{full}}} \approx 34.24\% \quad (4.4)$$

. The problem is worsened by the fact that samples shorter than the segment length defined in Section 5.4 are zero padded to the desired length adding even more silent parts.

As we don’t want our model to focus on generating silence a mask is generated for each sample specifying its silent ranges (cf. Figure 4.3). This mask is then used in the loss function to ignore the silent range (see Section 5.4).

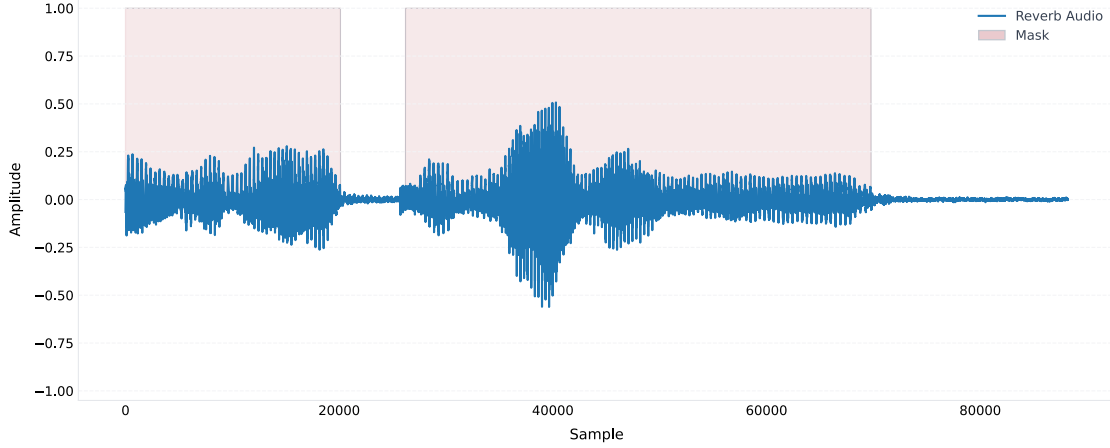


Figure 4.3: Signal with non-silent mask

4.2 Analysis of Applicable Loss Functions

Jonathan Kron

As described in Section 3.2.3.1, a loss function is a qualitative function that is used to objectively measure model performance by calculating the deviation of the model’s prediction from their ground truth counterpart. This deviation is mapped onto a real number that intuitively represents some error. To optimize model performance, this error must be minimized.

As shown in Section 3.2.3.4, different loss functions exist for different problem sets. Each research endeavor in machine learning must decide which loss function to use based on the nature of the problem, the data available, and the type of machine learning algorithm to be solved [Cia+24].

In the time or waveform domain, error-based regressive loss functions (e.g., MSE, SI-SNR, and PESQ) have identified themselves as well-performing in the field of dereverberation (see Chapter 2 and Section 3.3).

The metrics described in Section 3.3 can technically all be used as loss functions. The problem that all of them have in common is that none are specific to our task of dereverberation. It was shown that reverberation has the ability to either degrade or enhance a signal (cf. Section 3.1.2). Therefore, most metrics focus on signal corruptions that purely degrade sound quality (e.g., noise or sampling artifacts). In theory, PESQ appears to be a strong candidate, as it is a perceptual

metric that is invariant to both scaling and shifting, but as it is made for the evaluation of speech signals, effectiveness in diverse audio signals is doubtful. Rix et al. (2001) write: “Certain other applications have not yet been fully characterised or may need parts of the model to be changed. These include: music quality [...]”. An alternative lies in the Perceptual Evaluation of Audio Quality (PEAQ) model (cf. Section 3.3.5).

Other state-of-the-art measures include Google’s Virtual Speech Quality Objective Listener (ViSQOL) [Chi+20] as well as PEMO-Q [HK06] (cf. Section 3.3). But comparison indicates that PEAQ’s performance is not only competitive but sometimes even superior to newer approaches [DH20], meaning that ViSQOL and PEMO-Q were not further considered.

As explained in Section 4.1.2.1, the final dataset was reverberated offline using an implementation of the FreeVerb reverberator, allowing for export of size and wetness parameters on a per-sample basis. The wetness and size parameters are objective measurements, which we know to be true. A fully reverberated signal is defined as:

$$(\text{wetness} = 1) \wedge (\text{size} = 1) \tag{4.5}$$

. A fully dereverberated signal is defined as:

$$(\text{wetness} = 1) \vee (\text{size} = 1) \tag{4.6}$$

. This enables us to plot the different quality metrics against these objective measures and assess their applicability for the dereverberation task. Or, in other words, how well each metric estimates reverberation (and in turn dereverberation) of a signal.

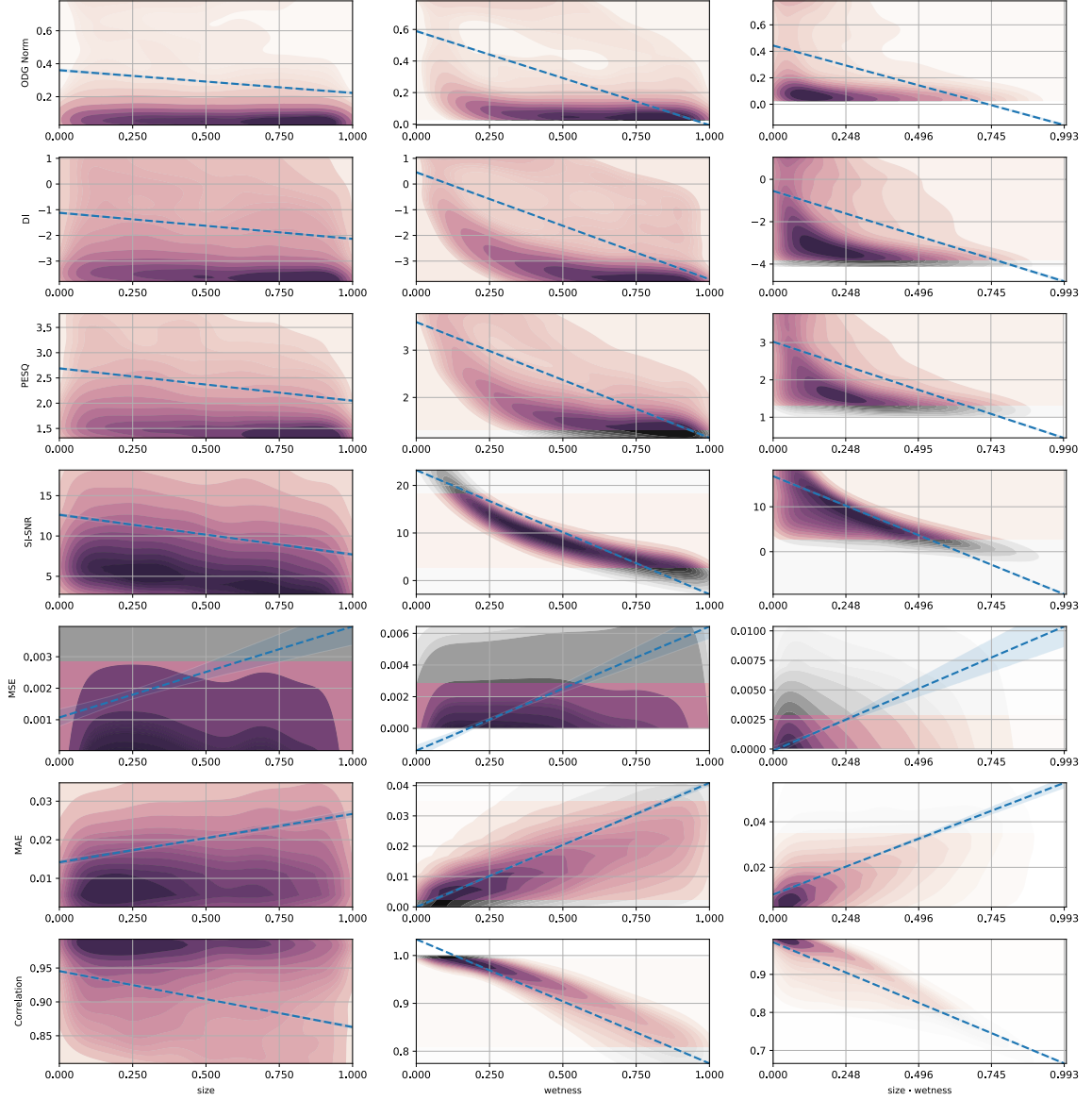


Figure 4.4: Metrics usable as loss functions analyzed over 16421 datapoints from test dataset (cf. Table 4.2), data between the 15th and 85th percentile is shown in color

Figure 4.4 shows a two-dimensional kernel density estimation (KDE) for each metric plotted against both the size and wetness parameters as well as size · wetness. The latter one was included as it is possible that samples with high size values simultaneously exhibit low wetness values and therefore are not reverberant (cf. Equation (4.5) and Equation (4.6)). The “size · wetness” plot corrects for that.

A kernel density estimation (KDE) plot is similar to a histogram but differentiates itself through a continuous density curve. This density curve was then

subdivided into 15 distinct plateaus or levels where contour lines were drawn. All data shown in color lies between the 15th and 85th percentile of data points, therefore excluding outliers. All data shown in grey is considered outlier data.

The dotted blue line represents a linear regression over all data points, including outliers. The light blue confidence interval band represents the 95% confidence interval for the regression line.

All tested signals were time and amplitude aligned. The only difference being the reverberation of the processed signal. Wetness and size values of the reverberator are selected randomly from a uniform distribution.

Prior to analysis, the ODG score was normalized

$$\text{ODG}_{\text{norm}} = \left(\frac{\text{ODG} + 4}{4} \right)^{\perp_1}_{\top_0} \quad (4.7)$$

. This procedure did neither aid nor hinder analysis but meant that the interpretation of the absolute value of the ODG score changes from “lower is better” to “lower is worse”. It was carried out as a remnant from early tests described in Section 4.3.

Figure 4.4 shows that ODG is not a good indicator of dereverberation performance, as most values stay between 0 and 0.2 in similar densities for the entire range of size and wetness values, meaning that most test signals, even those with little reverberation, were classified as annoyingly impaired.

The DI score was not normalized as the exact value range is unknown to us. It behaved similarly to the ODG score but showed slightly better performance against the wetness parameter but arguably worse performance against the size parameter, where many strongly reverberated signals are classified as “good”.

Although PESQ was only proven to work on speech signals, it showed a slightly improved performance compared to the DI score.

It is evident that the SI-SNR metric performs best as a judgment of dereverberation performance. The wetness KDE plot shows a strong correlation of absolute SI-SNR value and reverberation influence. As wetness and size values are

randomly sampled from a uniform distribution, the SI-SNR density stays mostly the same over the entire wetness range, which is the desired behavior. Although more outlier data is present in the size plot, a clear downward trend can be examined in the highest density parts of the KDE plot, further strengthening the assessment that SI-SNR predicts reverberation well in diverse audio signals.

The MSE metric shows neither meaningful predictive performance in the KDE plot nor a narrow confidence interval, with the wide spread undermining the regression line.

In similar fashion, the MAE shows poor performance against the size parameter. The wetness plot shows a slight increase of MAE against increasing wetness values. The problem is that this increase happens all below 0.04, meaning that every wetness value is assigned a very low error value or, in other words, is interpreted as “good”.

The correlation metric exhibits a similar problem where not only the size plot shows subpar performance, but the values in the wetness plot range from 1 to 0.8, which signifies a “very strong association” across the entire wetness range (cf. Section 3.3.2) making interpretation of the correlation value with respect to reverberation challenging. Furthermore, the density of the correlation-wetness plot does not align with the uniform distribution.

4.3 Perceptual Quality Network

Jonathan Kron

Although Section 4.2 shows the SI-SNR metric to have good qualities regarding the assessment of dereverberation performance in diverse audio signals according to the wetness parameter, the size parameter is not well represented. Calculating exact truths about a reverberated signal without the use of a neural network is nearly impossible, as it either requires knowledge of the sound source or the ability to model the reverb tail, which is not possible in short continuous utterances [Rat+03]. The SI-SNR, like all metrics introduced in Section 3.3, suffers from the need for a “golden” reference, which, as Fu et al. (2018) write “considerably restricts the practicality of such assessment tools [...]”. The presence of MOS

tests shows that humans can evaluate signal quality without the need of such a reference signal [Fu+18]. Motivated by these shortcomings, we introduce our own loss network initially coined “Perceptual Quality Network”.

We place the following requirements on this loss network. It must be differentiable and support fast inference as it is to be used as a loss function (see Section 3.2.3.1). As we plan to use it on a dataset of diverse audio signals (cf. Section 4.1.1), it must support wideband analysis up to 44.1 kHz.

Differentiability is given by the fact that the computations of a neural network are differentiable. The only exception is the activation function ReLU, which does not have a derivative in $z = 0$. In a real application the gradients are almost never zero, so this was ignored.

Similar in nature to Quality-Net [Fu+18], which estimates a PESQ score for a given signal (see Section 2.4), our initial idea was to estimate a combination of the ODG score as well as size and wetness parameters. Therefore, our model combines a perceptual and an objective approach.

Every training data pair consists of a reverberated audio sample serving as the network’s input, along with four target values the network is trained to predict: the normalized size and wetness parameters used to generate the reverberation, the normalized ODG score derived from a PEAQ comparison with the original sample, and a quality score defined as:

$$\text{quality} = \text{ODG}_{\text{norm}} \cdot (1 - \text{wetness}_{\text{norm}} \cdot 0.4) \cdot (1 - \text{size}_{\text{norm}} \cdot 0.3) \quad (4.8)$$

. This quality score is used as a loss in the dereverberation network (see Section 5.4).

Implementation details regarding model architecture and loss functions used to qualify ODG, size, wetness, and quality predictions are discussed in Section 5.2. Results are shown in Section 6.3, and an evaluation of the performance of the perceptual quality network is found in Section 7.3.

4.4 Objective Quality Network

Section 4.2 shows PEAQ as being a flawed metric to estimate dereverberation performance. This conclusion is supported by the findings in Section 7.3. Challenging the idea that a perceptual metric is best, we decided to test a purely objective network as a loss function.

Adopting the same structure as the perceptual quality network, the objective quality network only estimates the size and wetness attributes by defining the quality score as:

$$\text{quality} = 1 - 0.6 \cdot \text{wetness} - 0.4 \cdot \text{size} \quad (4.9)$$

Wetness was given more importance because Section 6.3 has shown the network as having better estimation performance for the wetness parameter.

Implementation details regarding model architecture and loss functions are discussed in Section 5.3. Results are shown in Section 6.3, and an evaluation of the performance is found in Section 7.4.

5 Implementation & Experimental Setup

This section details the implementation of our three proposed networks: the perceptual quality network, the objective quality network, and the dereverberation network. The state-of-the-art model Conv-TasNet required retraining and is therefore also covered in this context.

5.1 Conv-TasNet for diverse audio dereverberation Leo Kling

Conv-TasNet [LM19] operates in the time domain using a learned encoder–TCN–decoder architecture, where a temporal convolutional network estimates a multiplicative mask over the encoded signal to isolate a target source (cf. Section 2.1). Although originally designed for speech source separation, the masking paradigm is conceptually compatible with dereverberation. Late reflections overlap with the direct sound in the encoder representation, and a mask can in principle suppress this reverberant energy while retaining the direct component.

No pre-trained dereverberation weights were publicly available. Weights linked from the original repository were trained for speaker separation only and are thus not applicable to this task. Attempts to obtain suitable weights from the original authors received no reply. We therefore trained the model from scratch using the implementation linked in the paper².

The encoder is a 1-D convolution (512 channels, window 2 ms at 8 kHz). The TCN separator consists of 8 layers across 3 stacks with feature dimension 128 and depthwise-separable convolutions of kernel size 3. The decoder mirrors the encoder via a transposed convolution. The model is configured as a single-source system. The 8 kHz sample rate imposes a hard frequency ceiling of 4 kHz, which excludes upper harmonics and air that are perceptually important for music and broadband diverse audio content [ISO19, Pum50, Sha49].

The original training sets WSJ0-2mix and WSJ0-3mix [Gar+07] used in the Conv-TasNet paper are not publicly available, requiring an alternative training dataset. We used LibriSpeech [Pan+15], resampled to 8 kHz and segmented

²<https://github.com/naplab/Conv-TasNet>

into random 4-second crops per iteration. Reverberation is applied on-the-fly by convolving the dry signal with one of five impulse responses from the preprocessing pipeline (see Section 4.1.2.1), yielding time-aligned wet/dry pairs with varied room conditions across epochs. The training dataset is therefore speech-only, which biases the mask priors toward speech characteristics and is expected to reduce generalization to music and other diverse audio content.

Training used the Adam optimizer [KB17] with a learning rate of 10^{-3} , gradient clipping with a maximum L_2 -norm of 5.0 [LM19], and a batch size of 32 over 100 epochs via PyTorch Lightning. For the loss function, we initially followed the original paper and trained with the SI-SNR loss, but were unable to achieve meaningful convergence. We subsequently experimented with a multi-scale spectral (MSS) loss, which operates in the frequency domain and has shown promise in related audio tasks, but this too failed to yield satisfactory results. We ultimately trained with MSE loss directly on the time-domain waveform, which provided stable training dynamics and was used for all reported experiments.

5.2 Perceptual Quality Network

Jonathan Kron

The perceptual quality network was implemented twice. Section 5.2.1 shows the initial implementation of the perceptual quality network. It features a simple encoder network and prediction heads for each scoring metric.

A second implementation based on CNN14 as introduced by Kong et al. (2020) was written to address the shortcomings of the first implementation as mentioned in Section 7.3.1.

5.2.1 Simple CNN

The initial implementation of the perceptual quality network is based on a simple two-dimensional CNN. This architecture was chosen because CNNs have been widely adopted in audio machine learning [Gra+25] and have shown great performance. The small computational cost increase as compared to a waveform-domain approach was a nonissue, as this network was not to be used during inference but only during training of the dereverberation model (see Section 5.4).

The forward pass begins with conversion to a log-magnitude spectrogram via the STFT (logarithmic compression is discussed in Section 5.2.2), followed by a shared encoder consisting of three 2D convolutional layers, each with batch normalization, ReLU activation, and max pooling.

The output of this shared encoder is then fed into three prediction heads, each corresponding to one of the three initial labels (wetness, size, ODG). The output of each prediction head is concatenated with the shared encoder output and fed into the quality prediction head, giving this model the ability to predict all parameters at once.

This gives us a better chance at debugging (cf. Section 7.3.2) and the opportunity to use just one of the four predictions as a loss function. In the end, only the quality prediction was utilized.

Table 5.1: Architecture of the initial implementation of the perceptual quality network

Perceptual Quality Network (Initial)		
Log-magnitude spectrogram STFT: n_fft=2048, hop_length=512		
$\left(\begin{array}{c} 7 \times 7@32 \\ \text{BN, ReLU} \end{array} \right)$		
MaxPool 2×2		
$\left(\begin{array}{c} 5 \times 5@64 \\ \text{BN, ReLU} \end{array} \right)$		
MaxPool 2×2		
$\left(\begin{array}{c} 3 \times 3@128 \\ \text{BN, ReLU} \end{array} \right)$		
AdaptiveAvgPool (4×4)		
Flatten FC 2048 $\rightarrow$ 256, ReLU, Dropout(0.3)		
ODG Head FC 256 $\rightarrow$ 64 ReLU FC 64 $\rightarrow$ 1 Sigmoid	Size Head FC 256 $\rightarrow$ 32 ReLU FC 32 $\rightarrow$ 1 Sigmoid	Wetness Head FC 256 $\rightarrow$ 32 ReLU FC 32 $\rightarrow$ 1 Sigmoid
Concat [features(256) odg(1) size(1) wetness(1)] $\rightarrow$ 259		
Quality Head FC 259 $\rightarrow$ 64, ReLU FC 64 $\rightarrow$ 1, Sigmoid		

Table 5.1 shows the architecture of the initial implementation. The number after the “@” symbol indicates the number of feature maps. AdamW was used as an optimizer [LH19] with a learning rate of 10^{-3} and a weight decay value of 10^{-2} . A per-prediction-head loss was calculated using the MSE. The total loss was defined as:

$$\text{loss} = 2 \cdot \text{loss}_{\text{quality}} + \text{loss}_{\text{odg}} + 0.75 \cdot \text{loss}_{\text{size}} + 0.75 \cdot \text{loss}_{\text{wetness}} \quad (5.1)$$

5.2.2 CNN14

Compared to the initial implementation, this version offers a number of improvements. Mainly a new shared encoder architecture based on the CNN14 network

(cf. Table 5.2), which was introduced as a real-time audio pattern recognition model by Kong et al. (2020). We thought it fitting as we were trying to solve an adjacent problem (audio characteristic recognition) with good performance, as we did not want to needlessly slow down the training process of the dereverberation network.

A second improvement has been made in the spectrogram conversion. The log-magnitude spectrogram was replaced with a log-mel spectrogram, offering a perceptually oriented base for the encoder. The mel scale compresses higher frequencies more than lower ones. Therefore mimicking human perception of audio. The conversion from hertz into mels is defined as [O'S87]

$$m = 2595 \cdot \log_{10} \left(1 + \frac{f}{700} \right) \quad (5.2)$$

. The mel scale was chosen in favor of the bark scale as it is the most used and best-performing scale in computational acoustics (e.g., in ASR [Dho+19, Sim26]). Prediction heads as well as loss calculations were not subject to change.

The logarithmic compression of the mel scale (similar to the compression in Section 5.2.1) is motivated by the human perception of loudness. Logarithmic compression also helps in the representation of the mel value range for use in neural network training. Without any compression, the mel scale is condensed in a small range with extreme outliers. This means the network must be trained with higher numerical precision, hence making it more susceptible to noise. It was shown that decibel-scaled melspectrograms always outperform the linear versions [Cho+17].

Weights of the shared encoder are initialized using the Xavier uniform distribution as described in [GB10].

Table 5.2: Architecture of the CNN14 based implementation of the perceptual quality network

Perceptual Quality Network (CNN14)		
Log-mel spectrogram sr=44100, n_fft=2048, hop=512, n_mels=128		
BN (128 mel bins)		
$\left(\begin{array}{c} 3 \times 3@64 \\ \text{BN, ReLU} \end{array} \right) \times 2$		
AvgPool 2×2 , Dropout(0.2)		
$\left(\begin{array}{c} 3 \times 3@128 \\ \text{BN, ReLU} \end{array} \right) \times 2$		
AvgPool 2×2 , Dropout(0.2)		
$\left(\begin{array}{c} 3 \times 3@256 \\ \text{BN, ReLU} \end{array} \right) \times 2$		
AvgPool 2×2 , Dropout(0.2)		
$\left(\begin{array}{c} 3 \times 3@512 \\ \text{BN, ReLU} \end{array} \right) \times 2$		
AvgPool 2×2 , Dropout(0.2)		
$\left(\begin{array}{c} 3 \times 3@1024 \\ \text{BN, ReLU} \end{array} \right) \times 2$		
AvgPool 2×2 , Dropout(0.2)		
Global pooling (mean over freq.) Max + Mean pool over time $\rightarrow$ 1024-dim Dropout(0.5)		
FC 1024 $\rightarrow$ 512, ReLU, Dropout(0.3)		
ODG Head FC 512 $\rightarrow$ 128 ReLU, Dropout(0.3) FC 128 $\rightarrow$ 1 Sigmoid	Size Head FC 512 $\rightarrow$ 64 ReLU, Dropout(0.3) FC 64 $\rightarrow$ 1 Sigmoid	Wetness Head FC 512 $\rightarrow$ 64 ReLU, Dropout(0.3) FC 64 $\rightarrow$ 1 Sigmoid
Concat [features(512) odg(1) size(1) wetness(1)] $\rightarrow$ 515		
Quality Head FC 515 $\rightarrow$ 128, ReLU, Dropout(0.3) FC 128 $\rightarrow$ 1, Sigmoid		

5.3 Objective Quality Network

ODG was shown to worsen the performance of the perceptual quality network (see Section 7.3.2). Consequently, the perceptual quality network, as seen in Section 5.2.2, was adjusted to base the prediction only on the wetness and size parameters.

This change was done in two stages. Firstly, only the quality score was modified by removing the ODG score (cf. Section 4.4). Loss calculation was left unchanged. The second stage also changed the loss calculation from Equation (5.1) to just

$$\text{loss} = \text{loss}_{\text{quality}} \quad (5.3)$$

Both stages are evaluated in Section 7.4.

5.4 Dereverberation Network

Leo Kling

A first implementation of the dereverberation network was based on a fully generative approach, consisting of an encoder, a TCN, and a decoder as in Conv-TasNet but without the masking operation. The decoder was expected to generate the dereverberated signal directly from the TCN representation. However, this approach proved to be unfeasible with low computational cost, eventually requiring higher parameter counts. The model was able to overfit on the training data but did not generalize well to the validation set, as shown in Section 7.7. We therefore switched to a masking approach, more in line with the original Conv-TasNet architecture.

This model operates directly on waveforms. A reverberant input signal $\mathbf{x} \in \mathbb{R}^L$ (where L is the signal length) is first projected into a learned latent space by a strided 1-D convolution $\mathbf{W}_{\text{enc}}$, processed by a TCN that predicts a real-valued mask, and finally reconstructed by a transposed convolution $\mathbf{W}_{\text{dec}}$ (cf. Figure 5.1):

$$\tilde{\mathbf{x}} = \mathbf{W}_{\text{dec}}^{\top} * (\sigma(\text{TCN}(\mathbf{W}_{\text{enc}} * \mathbf{x})) \odot (\mathbf{W}_{\text{enc}} * \mathbf{x})) \quad (5.4)$$

The sigmoid-gated mask $\sigma(\dots) \in (0, 1)^{C \times T}$ suppresses reverberation in the encoder feature space before the decoder maps the filtered representation back to a waveform estimate $\tilde{x}$.

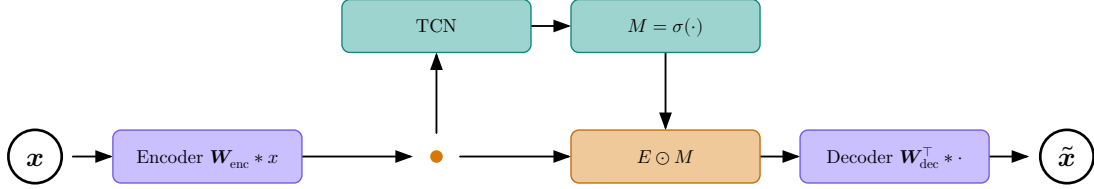


Figure 5.1: Forward pass of the dereverberation model.

As a loss function, the dereverberation network can either use the perceptual quality network or the SI-SNR. As shown in Section 4.1.2.3, a per-sample non-silent mask was generated. For both the SI-SNR and the quality network loss function, it is multiplied element-wise with both the ground truth and predicted signal.

Using the perceptual (see Section 4.3, Section 5.2.2) or objective (see Section 4.4, Section 5.3) quality network as loss functions requires little architectural change from the other metrics described in Section 4.2.

As the forward pass of a neural network is only a series of computations, all data requiring gradient calculation that is fed through the model will have the associated autograd graph built. To disable gradient calculation for the weights of the loss model, the gradient requirement for each parameter of the loss model is disabled. It is important to note that this should not be done using a global environment, as then all gradient calculations, including those of the dereverberation network’s weights, will cease.

As explained in Section 5.2.1, only the quality prediction was used. The final loss calculation for the dereverberation network is defined as:

$$\text{loss} = 1.0 - \text{quality} + \alpha \cdot \text{MSE}_{\text{loss}} \quad (5.5)$$

where quality is the predicted quality score described in Section 4.3, α is some factor between 0 and 1, and MSE_{loss} is the MSE between s and $\hat{s}$ (cf. Section 3.3.1).

Table 5.3: Architecture of the dereverberation network (N=encoder channels, X=num blocks per repeat, R=num repeats)

Dereverberation Network (TCN v4)		
Waveform input sr=44100, segment length = 4 s		
Encoder	Separator (TCN)	Decoder
1-D Conv encoder $N = 512$, win=2 ms	24 residual TCN blocks channels=179, kernel size=3	1-D transposed Conv decoder waveform reconstruction
$E = \mathbf{W}_{\text{enc}} * \mathbf{x}$	Dilations: $[1, 2, 4, 8, 16, 32, 64, 128] \times 3$ $X = 8, R = 3$	$\tilde{\mathbf{x}}$
Masking path	$M = \sigma(\text{TCN}(E))$	$\hat{E} = E \odot M$
Non-causal, layer_norm, ReLU, dropout=0.0, lookahead=0, skip connections enabled		

As shown in Table 5.3 our dereverberation network follows the core Conv-TasNet principle of encoder–masking–decoder processing but is adapted to a different task and operating regime than the original model by Luo and Mesgarani (2019). In contrast to the speech-separation setting of Conv-TasNet (multi-speaker mixtures at 8 kHz with permutation-invariant SI-SNR optimization), our configuration targets single-source dereverberation on paired wet/dry signals at 44.1 kHz. The structural backbone remains comparable, including the dilation pattern with $X = 8$, $R = 3$, and kernel size 3, while key capacity choices are shifted for broadband dereverberation. The encoder width is kept at $N = 512$ with a 2 ms analysis window, but the TCN channel width is reduced to 179, as shown in Table A.1 (derv_tcn_v4), to align with Conv-TasNet’s 5.1 million parameter count. In addition, optimization is not restricted to SI-SNR but is evaluated with both SI-SNR and perceptual/objective quality-based losses, reflecting the broader quality criteria required for diverse audio content.

6 Results

This chapter presents the results obtained from all evaluated models: the two state-of-the-art baselines Conv-TasNet and StoRM, the perceptual and objective quality networks, and the dereverberation network. The discussion covers performance on speech dereverberation, diverse audio dereverberation, and, in the case of the quality networks, their ability to indicate dereverberation performance.

6.1 Conv-TasNet

Leo Kling

As explained in Section 5.1, we trained a Conv-TasNet model from scratch on the LibriSpeech `train-clean-100` split, using the original Conv-TasNet architecture [LM19] and training procedure, but with MSE instead of SI-SNR as the loss function.

Three loss functions were evaluated in total. The SI-SNR, which serves as the original Conv-TasNet training objective, did not converge: the loss remained negative throughout and continued to decrease without producing usable predictions, with the best checkpoint reaching a validation SI-SNR of -69.72 dB after 118 epochs (cf. Figure 6.1). A MSS loss likewise showed convergence but only at an unreasonably high value, reaching a validation loss of 165,861.84 after 119 epochs (cf. Figure 6.2). This could be attributed to some configuration choices that were set implicitly and thus often fail to provide informative gradients as claimed by Schwär and Müller (2023).

Switching to a standard MSE loss resolved the issue: training converged stably to a validation loss of approximately 0.0009 after 125 epochs (cf. Figure 6.1). This is further discussed in Section 7.5.

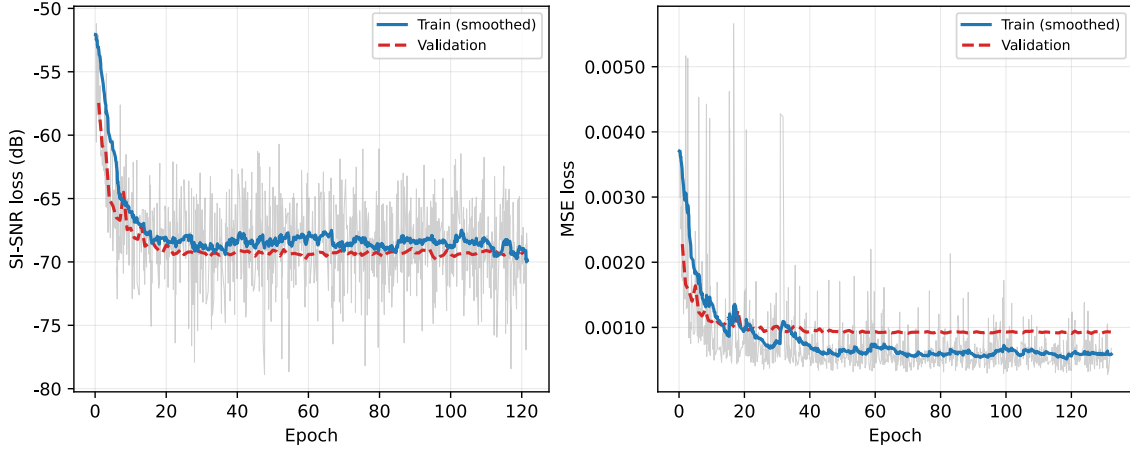


Figure 6.1: Training curves for Conv-TasNet with different loss functions. Smoothed using an exponential moving average with $\alpha = 0.05$.

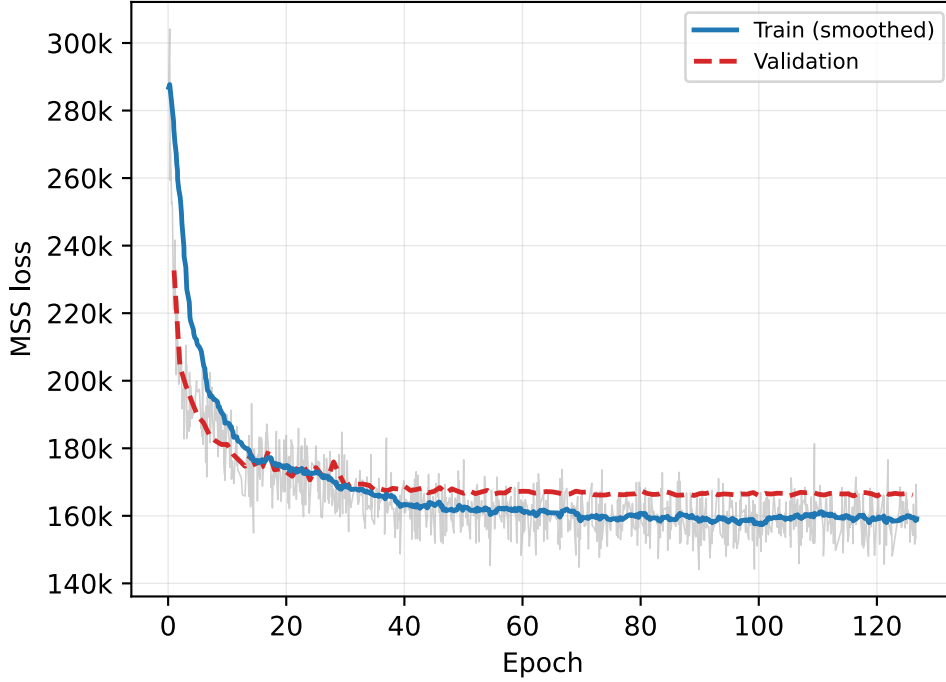


Figure 6.2: Training curve for Conv-TasNet with MSS loss. Smoothed using an exponential moving average with $\alpha = 0.05$.

The MSE-trained model was evaluated on the LibriSpeech **test-clean** split as well as on a diverse random subset taken from AudioSet covering speech, music, vehicles, and environmental sounds.

On speech samples, the model reduces reverberation tails and produces audible dereverberation. It can also be observed that the model applies a low-

pass filter, reducing high frequencies above about 2.5 kHz by about 20 dB (see Figure 6.3).

Detailed results of the evaluation done on the MSE-trained model can be seen in Table 6.1. A positive SI-SNR indicates dereverberation, while the PESQ score still indicates a “*poor*” performance (cf. Section 3.3.4).

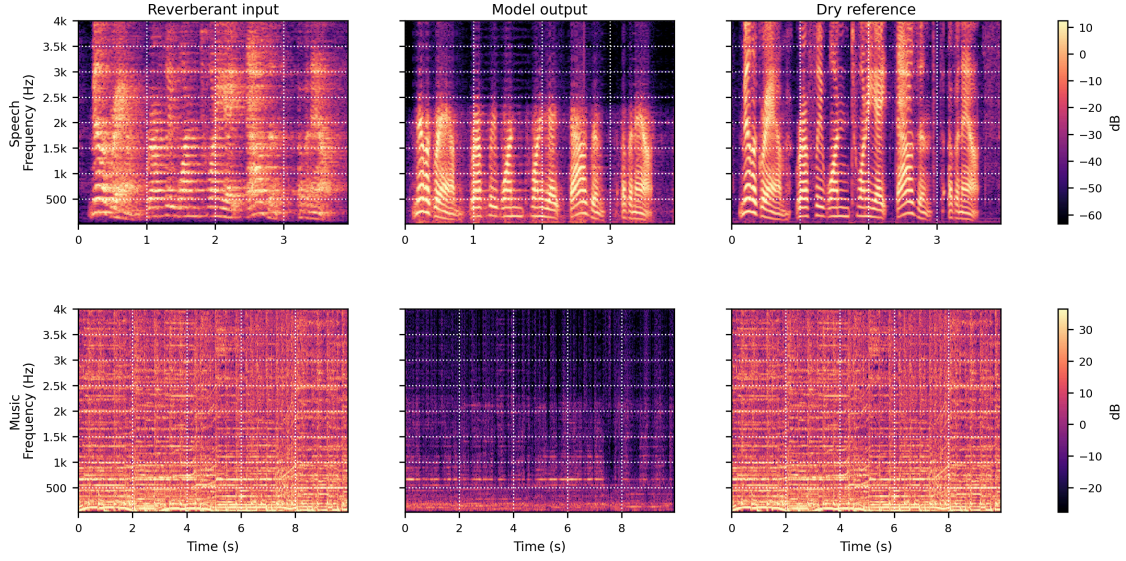


Figure 6.3: Spectrogram comparison of input (left) and output (middle) of the MSE-trained Conv-TasNet on a speech (top) and music (bottom) sample.

On music and non-speech content, however, the model introduces noticeable timbral artifacts.

Table 6.1: Conv-TasNet dereverberation metrics on LibriSpeech `test-clean` ($N = 2620$)

Metric	Mean	Std	Min	Max
SI-SNR	10.036	3.856	-6.669	19.226
PESQ	1.726	0.304	1.117	3.015
WV-MOS	1.695	0.441	1.229	3.350

These limitations – the 4 kHz bandwidth ceiling, speech-only training data, and the uncertainty of SI-SNR as a training objective for diverse audio – motivate the development of a dedicated dereverberation model trained on broadband diverse content and supported by a perceptual loss network.

6.2 StoRM

Leo Kling

Unlike Conv-TasNet, StoRM was not trained from scratch. We used the official pretrained dereverberation checkpoint provided by the authors, trained on the WSJ0 corpus reverberated with the REVERB challenge dataset [Gar+07, Kin+13, Lem+23]. The training data consists of speech recordings sampled at 16 kHz, establishing an 8 kHz frequency ceiling. Architecturally, StoRM follows a generative stochastic regeneration approach: a discriminative denoiser first produces an initial estimate of the clean signal, which a score-based diffusion model then refines through a learned reverse process [Lem+23]. This contrasts with Conv-TasNet’s discriminative masking, and the iterative inference required by the diffusion component has direct negative implications for computational cost.

On in-domain speech signals, StoRM achieves strong dereverberation quality. Table 6.2 shows the evaluation from the original paper. These numbers serve as an upper bound for speech dereverberation quality achievable with this model.

Table 6.2: StoRM dereverberation metrics on the WSJ0+REVERB test set, reproduced from Lemerrier et al. (2023).

Method	WV-MOS	PESQ	ESTOI	SI-SDR	SI-SIR	SI-SAR
Mixture	1.78 ± 0.99	1.36 ± 0.19	0.46 ± 0.12	-7.3 ± 5.5	-7.5 ± 5.4	—
SGMSE+	3.49 ± 0.39	2.66 ± 0.45	0.85 ± 0.06	2.4 ± 7.2	11.6 ± 9.9	2.8 ± 6.8
NCSN++	2.99 ± 0.38	2.08 ± 0.47	0.85 ± 0.06	6.1 ± 3.8	21.4 ± 7.0	6.1 ± 3.7
GaGNet	2.40 ± 0.52	1.59 ± 0.37	0.68 ± 0.09	-0.5 ± 4.8	7.7 ± 4.0	0.2 ± 5.1
StoRM	3.73 ± 0.32	2.83 ± 0.42	0.88 ± 0.04	6.5 ± 4.0	22.9 ± 8.2	6.5 ± 3.9

In our own listening tests on speech samples, this quality is confirmed: reverberation tails are cleanly removed with rarely any audible artifacts (see Figure 6.4). Informally, the model appears to perform slightly worse on female voices, which may be attributable to a gender bias in the WSJ0 training corpus toward male utterances, even though the authors claim: “[...] about half the speakers are male and half female” [Gar+07]. Compared to Conv-TasNet, StoRM produces a markedly wider frequency response up to 8 kHz, avoiding the strong low-pass filtering effect observed in the MSE-trained Conv-TasNet output.

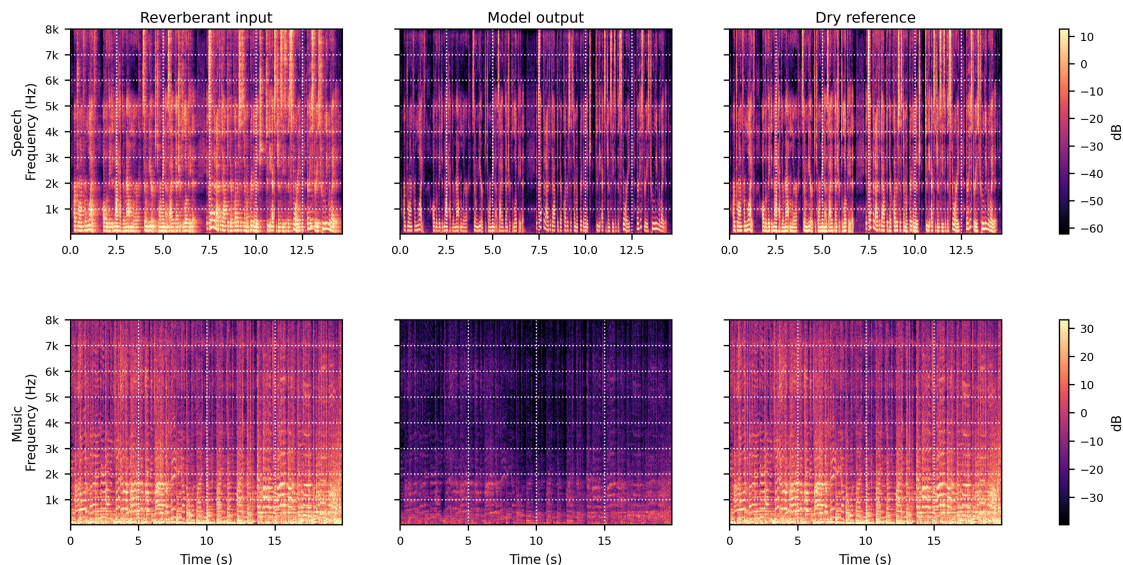


Figure 6.4: Spectrogram comparison of input (left) and output (middle) of StoRM on a speech (top) and music (bottom) sample.

On music and other non-speech content, the model’s behavior is less predictable. As the pretrained checkpoint has no exposure to non-speech signals during training, generalization is limited to the extent that spectral patterns of broadband audio are covered by the speech-domain prior. In listening tests, music samples processed by StoRM tend to exhibit subtle timbral changes compared to the unprocessed input, without achieving a consistent reduction of the reverberant tail. This out-of-domain degradation is expected given the training data composition and is explored further in the quantitative comparison in the next section.

The iterative reverse diffusion inference requires many sequential neural network evaluations per sample, making StoRM substantially more expensive than Conv-TasNet. On a single H100 GPU (CLAIX-2023-ML), processing 2048 AudioSet samples took 6 h 14 m 51 s, compared to 4 m 15 s for Conv-TasNet (cf. Table 7.4). Real-time application of this pretrained model is therefore not feasible without architectural modifications such as reducing the number of reverse diffusion steps or distillation.

6.3 Perceptual Quality Network

Jonathan Kron

All results presented in the following section are based on the dataset as described in Section 4.1.1. Results regarding prediction performance as well as use as a loss function are discussed. Results regarding the impact of the perceptual quality network on the dereverberation network are demonstrated in Section 6.5. Prediction performance was analyzed using the MAE, MSE and correlation (cf. Section 3.3). The average over the entire testing subset (cf. Table 4.2) is depicted in Table 6.3. As discussed in Section 7.3.1, all perceptual quality network experiments were conducted using the CNN14-based implementation. Therefore no results of the simple CNN, as detailed in Section 5.2.1, are shown going forward.

Table 6.3: MAE, MSE and correlation of all perceptual quality metrics at epoch 195

Type	MSE	MAE	Correlation
quality	0.005127	0.042432	0.966335
odg	0.006775	0.048189	0.968081
size	0.03663	0.156579	0.910949
wetness	0.011141	0.088021	0.982811

To visualize prediction performance, KDE plots, as seen in Figure A.1, were generated. Each prediction-head output (quality, wetness, size, ODG) is plotted against its ground truth counterpart for each data pair of the testing subset. Akin to the testing done in Section 4.2, the quality score, which was used as the loss function of the dereverberation network (see Section 5.4), was plotted against the real size, wetness, and size $\cdot$ wetness values of the testing data (cf. Figure 6.5). Further evaluation of these plots is found in Section 7.3.2.

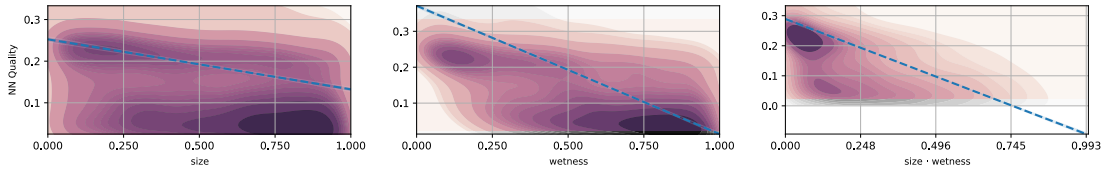


Figure 6.5: Quality score prediction of the perceptual quality network at epoch 195 analyzed as loss function over 16421 datapoints from the testing subset (cf. Table 4.2). Data between the 15th and 85th percentile is shown in color.

6.4 Objective Quality Network

Jonathan Kron

As described in Section 5.3, the objective quality network underwent a two-stage refinement process. Retraining for 61 epochs using the updated quality score resulted in the metric averages as seen in Table 6.4 and the prediction versus ground truth plots as seen in Figure A.2.

Table 6.4: MAE, MSE and correlation of all objective quality metrics using the updated quality score at epoch 61

Type	MSE	MAE	Correlation
quality	0.014664	0.110005	0.961182
odg	0.00929	0.06119	0.958197
size	0.023378	0.12407	0.908911
wetness	0.02126	0.129864	0.972678

Retraining for 166 epochs using the updated quality score as well as the updated loss function calculation resulted in metric averages as seen in Table 6.5 and prediction versus ground truth plots as seen in Figure A.3.

Table 6.5: MAE, MSE and correlation of all objective quality metrics using the updated quality score and loss at epoch 166

Type	MSE	MAE	Correlation
quality	0.01546	0.112525	0.96009
odg	0.495343	0.628064	-0.531829
size	0.110419	0.276881	0.472437
wetness	0.15364	0.308015	0.276411

It must be noted that with the updated loss function calculation, which only uses the predicted quality score to adjust the model’s weights, the results in Table 6.5 and Figure A.3 for size, wetness, and ODG become meaningless.

Plotting the objective quality network as seen in Table 6.5 against the corresponding size, wetness, and size · wetness values resulted in the plot seen in Figure 6.6. This visualization is discussed in Section 7.4.

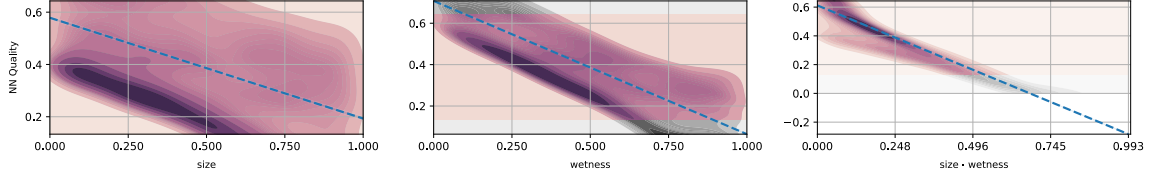


Figure 6.6: Quality score prediction of the objective quality network using the updated quality score and loss at epoch 166 analyzed as loss function over 16421 data points from the testing subset (cf. Table 4.2). Data between the 15th and 85th percentile is shown in color.

Analyzing the inference speed of the objective quality network, it was found that for a total length of 18.2 hours of audio data sampled at 44.1 kHz, the total length of inference time amounted to about 7.74 seconds. As the objective quality network shares the same architecture with the perceptual quality network, this result can be assumed similar for all of the above tested configurations.

6.5 Dereverberation Network

Leo Kling

As described in Section 5.4, the dereverberation network was trained using multiple loss functions. Namely, SI-SNR as well as the perceptual quality network (Section 4.3) and objective quality network (Section 4.4) were tested.

First, a baseline using the SI-SNR was established. Training was done over 62 epochs with early stopping at epoch 55, resulting in a validation loss of 16.6097 dB (cf. Figure 6.7). Table 6.6 is showing performance metrics for model outputs and is further discussed as part of the evaluation in Section 7.7.

Table 6.6: Evaluation metrics for the SI-SNR baseline model (epoch 55, validation loss = 16.6097 dB). PEAQ metrics computed on $n = 1758$ samples.

Metric	Baseline	Enhanced	Δ
SI-SNR (dB)	9.98 ± 8.67	18.78 ± 7.01	$+8.80 \pm 3.02$
PESQ	1.88 ± 0.84	2.94 ± 0.78	$+1.06 \pm 0.38$
PEAQ ODG	-3.57 ± 0.75	-3.65 ± 0.34	-0.11 ± 0.55
PEAQ DI	-2.92 ± 1.41	-2.85 ± 0.92	$+0.01 \pm 1.00$
MSE	0.0010 ± 0.0031	0.0216 ± 0.0233	-0.0206 ± 0.0217

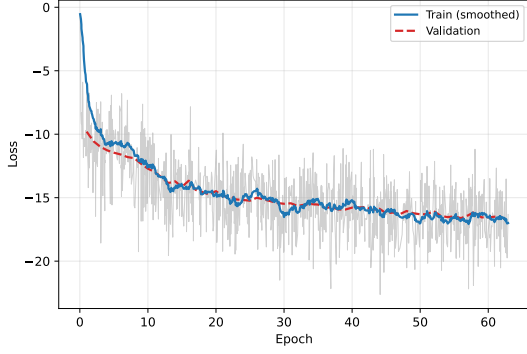


Figure 6.7: Loss curve of the dereverberation network trained with SI-SNR loss. Smoothed using an exponential moving average with $\alpha = 0.05$.

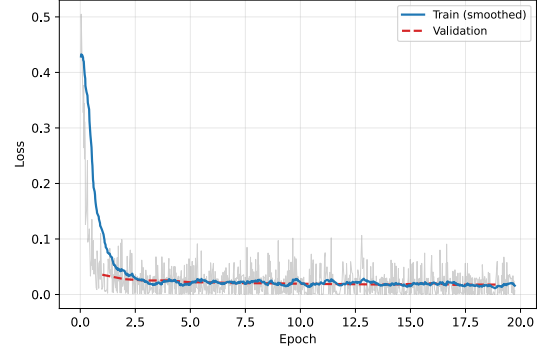


Figure 6.8: Loss curve of the dereverberation network trained with the objective perceptual network. Smoothed using an exponential moving average with $\alpha = 0.05$.

Results of the model output can be seen in Figure 6.9 and Figure 6.10, where Figure 6.9 is an earlier checkpoint at 16 epochs and a validation loss of -14.3848 dB.

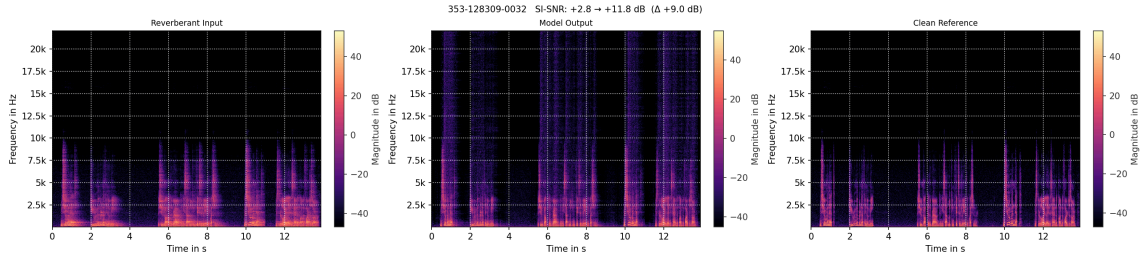


Figure 6.9: Spectrogram comparison of input (left) and output (middle) of the dereverberation network trained with SI-SNR loss at epoch 16 (validation loss = -14.3848 dB). The rightmost plot shows the original clean signal for reference.

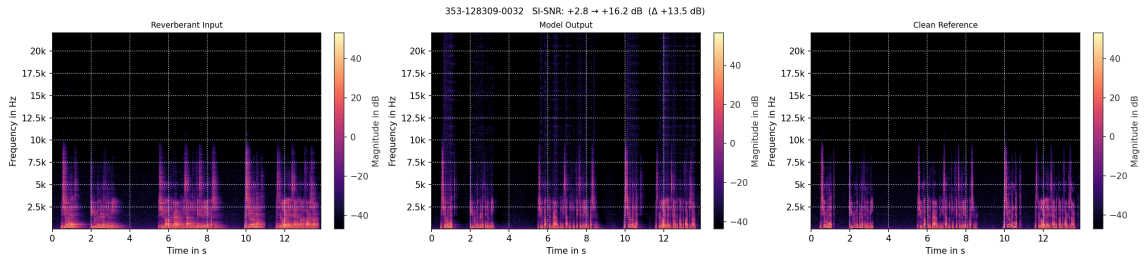


Figure 6.10: Spectrogram comparison of input (left) and output (middle) of the dereverberation network trained with SI-SNR loss at epoch 55 (validation loss = -16.6097 dB). The rightmost plot shows the original clean signal for reference.

Both the perceptual quality network and objective quality network were not successful in their application. Training and evaluation loss did both approach 0 (see Figure 6.8 for objective quality network) but showed no effective dereverberation.

In fact, both failed to reconstruct the original signal properly, resulting in audible artifacts and visible coloration of the source material (cf. Figure 6.11).

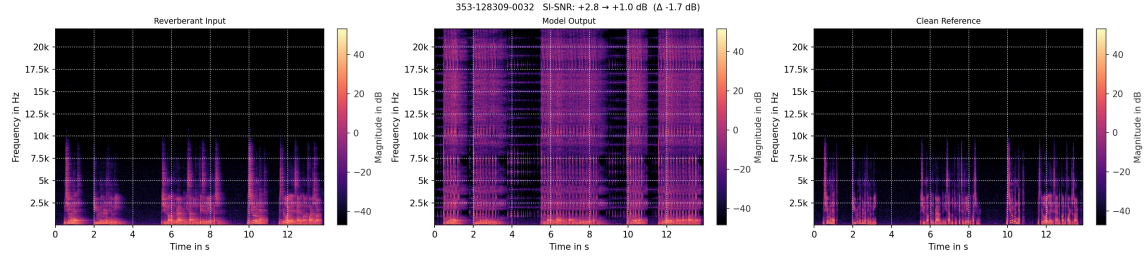


Figure 6.11: Spectrogram comparison of input (left) and output (middle) of the dereverberation network trained with the objective perceptual network at epoch 70 (validation loss = 0.0154).

The rightmost plot shows the original clean signal for reference.

7 Evaluation

This chapter discusses the results shown Chapter 6 in regards to the problem statement found in Section 1.2. Shortcomings of our approach introduced in Chapter 4 are debated and further research questions are asked.

7.1 Reverberation Time Distribution in the Dataset

The distribution of generated reverberation times (RT_{60}) for the samples in our dataset (see Section 4.1.2.1) is shown in Table 7.1 and Figure 7.1. These times were not measured on the reverberated signals directly but measured on an impulse that was reverberated with the same parameters as the audio sample. Measurements were then done using the Schroeder method [Sch68], which is based on the energy decay curve of the reverberant signal.

Table 7.1: RT_{60} distribution of reverberated dataset

Split	Mean [s]	Median [s]	Std [s]	Min [s]	Max [s]
All	1.480289	1.185601	0.787817	0.578050	2.805147
Train	1.481210	1.185624	0.788235	0.578050	2.805147
Val	1.475887	1.188571	0.784935	0.578050	2.805147
Test	1.480334	1.180454	0.788687	0.578050	2.805125

The histogram in Figure 7.1 shows the distribution of RT_{60} values across the entire reverberated dataset, with a mean of 1.48 s. Overall, it shows a wide variety of reverberation times while still being in a realistic range. The spike at the right end of the histogram is not expected, as wetness and size parameters were uniformly distributed, meaning that a “maximum” reverberation time should not be more common than any other reverberation time. This could be caused by a non-linear relationship between the wetness and size parameters and the resulting RT_{60} value, which could lead to certain combinations of parameters producing similar RT_{60} values more frequently. Another possibility is that there is a bug in the code that generates the reverberated signals or in the code that measures the RT_{60} values, which could lead to incorrect measurements or an overrepresentation of certain RT_{60} values. Further investigation would be needed to determine the exact cause of this spike in the histogram.

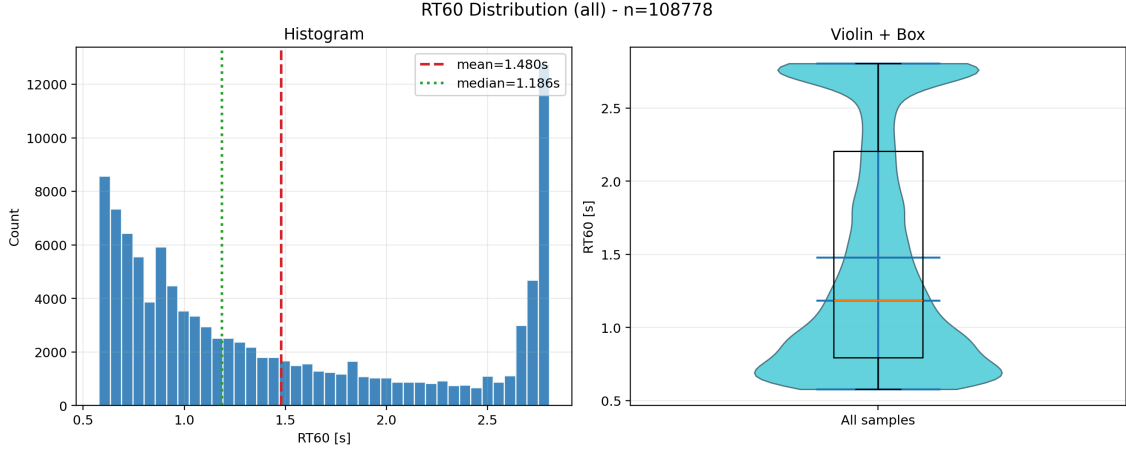


Figure 7.1: Distribution of RT_{60} values across entire reverberated dataset

7.2 Analysis of Applicable Loss Functions

The analysis of applicable loss functions carried out in Section 4.2 bases itself on the assumption that the size and wetness parameters that were used to parametrize the reverb are good indicators of dereverberation performance. It is reasonable to assume that wetness is a good indicator, as it is used as a factor to multiply the wet signal in the Freeverb implementation. Therefore it has no impact on the reverb tail itself, only making it quieter, which is what we expect of our dereverberation model. The size parameter behaves differently. It indirectly adjusts the intensity of each delayed signal of the comb filters. It is unclear how well a reduction in room size corresponds with reducing reverberation.

Figure 4.4 shows every metric performs worse against the size parameter. This might not be the metric’s fault; consequently, all conclusions based on the size plots alone require further investigation.

7.3 Perceptual Quality Net

Jonathan Kron

Evaluation of the perceptual quality network is done both for the initial simple implementation and the CNN14-based one. The evaluation of the simple CNN (see Section 7.3.1) focuses on the comparison of the two approaches, examining the improvement the CNN14-based implementation made. Section 7.3.2 discusses

the general applicability of the perceptual quality network as a loss function and its predictive performance compared to other metrics like the SI-SNR.

7.3.1 Simple CNN

This evaluation was carried out using a subset of the dataset discussed in Section 4.1.1. Both the simple CNN and the CNN14-based implementation were trained on the same subset. Training was done over 10 epochs, saving only the best epoch of the training process.

The output of all prediction heads is compared to their ground truth counterpart using MSE, MAE and correlation as comparative metrics. Table 7.2 shows absolute differences between ground truth and prediction as well as relative improvements between the simple CNN and the CNN14-based implementation for the quality score.

Table 7.2: Metric comparison of the quality score, taken from the best epoch of the first 10, between the simple CNN and CNN14 implementation

Metric	Simple CNN	CNN14	Relative improvement
MSE	0.00867	0.00587	32.31 %
MAE	0.04722	0.03964	16.06 %
Correlation	0.86715	0.91155	5.12 %

Table 7.2 shows improvements over all comparative metrics. Similar advancements have been made across all parameters (cf. Table 7.3).

Table 7.3: Relative improvement from the CNN14 implementation as compared to the simple CNN implementation over all metrics and parameters, taken from the best epoch of the first 10

Metric	quality score	ODG	size	wetness
MSE	32.31 %	36.19 %	35.97 %	36.07 %
MAE	16.06 %	14.6 %	25.28 %	25.32 %
Correlation	5.12 %	4.06 %	6.71 %	6.85 %

These findings motivated us to proceed with the CNN14-based implementation as described in Section 5.2.2 for all further experiments.

7.3.2 CNN14

The MSE, MAE and correlation averages shown in Table 6.3 indicate very good prediction performance of every prediction head on unseen data. Compared with Quality-Net (see Section 2.4), we achieved an ODG MSE value of 0.006775, where the ODG was normalized, while Quality-Net showed a PESQ MSE of 0.1266, with PESQ values ranging from 0 to 4. Dividing Quality-Net’s MSE by 4 results in a normalized MSE of 0.03165, which is still notably worse than ours.

Figure A.1 shows that the quality score as proposed in Section 4.3 is very dependent on the ODG. This turned out to be a problem, as ODG values were unexpectedly overrepresented in the 0 to 0.2 range, meaning that most signals were classified as imperceptibly impaired. In theory this shouldn’t impact the use of the network as a loss function, but some signals were graded significantly worse (in the 0.8 to 1 range), making gradient descent based on the ODG unpredictable. Wetness and size predictions behaved as expected with some skewing in the upper range (0.9 to 1). This problem can be observed in both plots, although a stronger impact is noticed in the size predictions. This observation led to the stronger weighting of the wetness prediction in Section 4.4.

Figure 6.5 plots the quality score against size, wetness, and size · wetness. This visualization has the same shortcomings as the ones in Section 4.2, as explained in Section 7.2. It is shown that compared with the ODG, the perceptual quality network has improved reverberation prediction performance. Unfortunately, it cannot compete with the SI-SNR (cf. Figure 4.4).

Section 4.1.2.3 noted that about one-third of our dataset was silence. It was therefore important to investigate whether model performance suffers from prolonged silence in an audio signal. To evaluate this, a single audio file was loaded, and quality, size, and wetness predictions were made. Iteratively, parts of the audio signal were replaced with zeros. Figure 7.2 shows the predicted values with increasing zero percentage. Remarkably, the model’s prediction kept steady up to about 90% zero percentage. It was therefore concluded that using the perceptual

quality network as a loss function on signals with silent parts should not pose an issue.

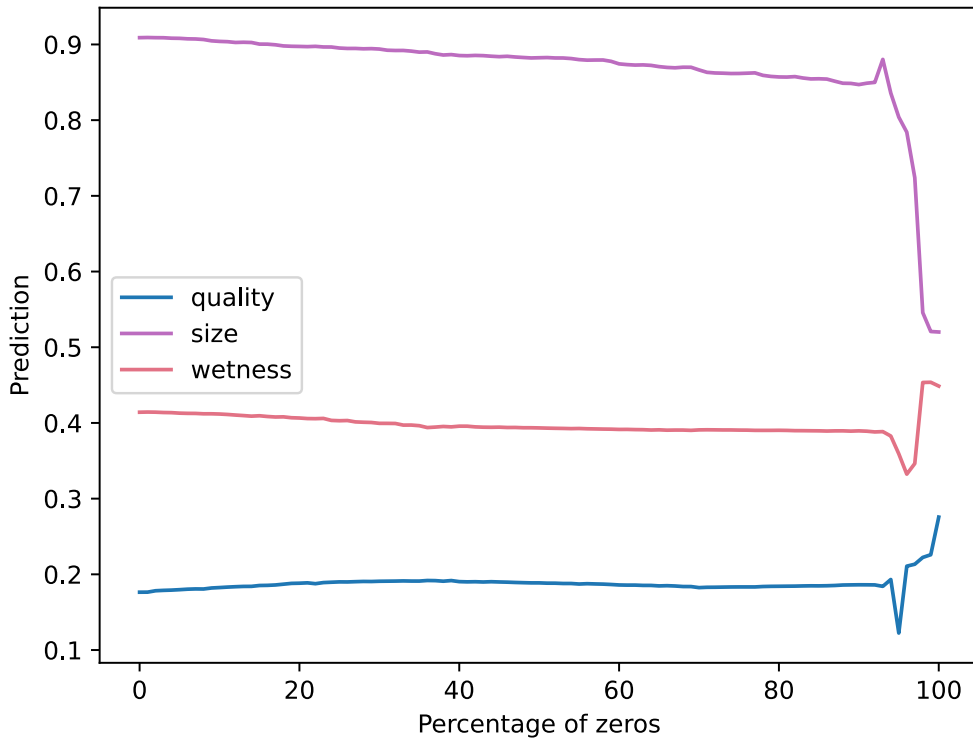


Figure 7.2: Prediction quality of perceptual net from signal with increasing zero percentage

Multiple shortcomings of the general approach to the training and using as a loss function of the perceptual quality network were identified. As described in Section 4.1.1, our dataset is in parts composed of the AudioSet and FSD50K. The samples found in these datasets are not necessarily recorded in anechoic conditions, meaning that it is possible that an already reverberated file was reverberated with a wetness of 0 and therefore mislabeled. As the FSD50K dataset is taken from a sample sharing site, it is less likely this problem has occurred there. A related issue concerns the training concept, as we used the same dataset for both the training of the perceptual quality network and the dereverberation network. While both used the same subsets (cf. Table 4.2), it is possible that the perceptual quality network wrongly overfitted on some samples. Further research should also investigate the

use of the activation function ReLU, as differentiability is not guaranteed (cf. Section 4.3).

7.4 Objective Quality Net

The MSE, MAE and correlation averages of the quality score shown in Table 6.4 demonstrate worse performance than the ones of the perceptual quality net (cf. Table 6.3). This is to be expected, as now all values of the quality score are uniformly distributed in the range of 0 to 1.

As we did not utilize singular ODG, size, and wetness predictions, it was decided that the extra computational resources used to propagate errors stemming from these prediction heads can be saved. Using the updated loss function (cf. Section 5.3), the metric averages of the quality score did not improve notably (cf. Table 6.5).

Figure 6.6 shows that the quality score stemming from the objective quality network using the updated loss function is competitive with the SI-SNR metric for indicating dereverberation performance (cf. Section 4.2). Predictions against the size parameter have improved substantially, exhibiting markedly reduced error. While predictions against the wetness parameter remain more dispersed, the non-linear curvature previously observed in the SI-SNR visualization (cf. Figure 4.4) is no longer present, indicating that optimization with respect to this metric should yield more stable and predictable convergence behavior.

Inference speed at 7.74 seconds for 18.2 hours of audio data sampled at 44.1 kHz means that using this network as a loss function should not pose any significant drawbacks in regard to training speed.

None of the shortcomings as identified in Section 7.3.2 were addressed or mitigated in any meaningful capacity.

7.5 SI-SNR Calculations

The SI-SNR values reported in Figure 6.1 and Table 7.4 warrant closer scrutiny. Specifically, when our Conv-TasNet re-implementation was trained using SI-SNR loss, convergence did not proceed as expected, with the loss failing to decrease

in a manner consistent with the improvements observed under MSE training. This raises the question of whether the SI-SNR calculations used for that training were correctly implemented and whether this accounts for the unexpected training behaviour.

To validate the implementation, the MSE-trained Conv-TasNet checkpoint was evaluated on the LibriSpeech test set and yielded an SI-SNR of 10.0 dB. This falls short of the 15.3 dB reported in the original Conv-TasNet paper [LM19], a gap that is too large to attribute solely to differences in training data or hyperparameters. While the model was not trained under identical conditions to the original work, the magnitude of the discrepancy suggests that a systematic issue may be present in the SI-SNR calculation. It should be noted that while the MSE checkpoint does fall short in SI-SNR compared to the original paper, it still demonstrates a significant improvement over the reverberant input, indicating that the model is learning to dereverberate to some extent.

7.6 Comparison of Conv-TasNet and StoRM for diverse signals

Leo Kling

Both Conv-TasNet and StoRM were trained exclusively on speech recordings and had no exposure to music, environmental noise, or other non-speech content. To assess how each model generalizes outside this group, both were applied to 2048 randomly sampled AudioSet clips spanning a wide range of acoustic scenes and event categories. Table 7.4 summarizes these metrics, while the full distributions are shown in Figure 7.3.

Table 7.4: Comparison of different metrics for the evaluation of dereverberation performance of diverse audio samples, evaluated on 2048 random AudioSet samples. The runtime is measured on a single H100 GPU (CLAIX-2023-ML) [CLA26] with 2048 Random AudioSet Samples.

Network	StoRM	Conv-TasNet
MSE	0.02996 ± 0.048	0.02795 ± 0.044
SI-SNR (dB)	-31.39 ± 11.71	-31.26 ± 18.6
PESQ-WB	1.35 ± 0.29	1.45 ± 0.44
PESQ-NB	1.82 ± 0.4	1.81 ± 0.64
ODG	-3.67 ± 0.48	-3.83 ± 0.14
DI	-3.14 ± 1.06	-3.5 ± 0.59
Runtime (hh:mm:ss)	06:14:51	00:04:15
Runtime per sample (s)	10.98	0.12

Across all metrics, both models perform substantially below their in-domain speech statistics. PESQ-WB reaches only 1.35 (StoRM) and 1.45 (Conv-TasNet), far below StoRM’s in-domain speech result of 2.83 (Table 6.2). ODG values of -3.67 and -3.83 place both models near the lower end of the five-step degradation scale, indicating consistently “annoying” to “very annoying” perceived quality. The boxplots in Figure 7.3 confirm that these results are not driven by a few extreme samples: distributions are broad but consistent, with no single extreme point pulling results in one direction. Boxplots for SI-SNR and PESQ show some narrower interquartile ranges and shorter whiskers for the StoRM model, meaning Conv-TasNet’s performance is more inconsistent. A recurring informal observation from listening tests and viewing spectrograms is that both models tend to lower the output level relative to the input, especially for non-speech signals. This unintended effect is visible in the spectrograms (Figure 6.3 and Figure 6.4).

It should be noted that the SI-SNR values obtained from our Conv-TasNet implementation trained with the MSE loss function may deviate from those reported by the original implementation (see Section 7.5), although a comparable disparity between in-domain and out-of-domain performance can be anticipated when using the original implementation under analogous conditions.

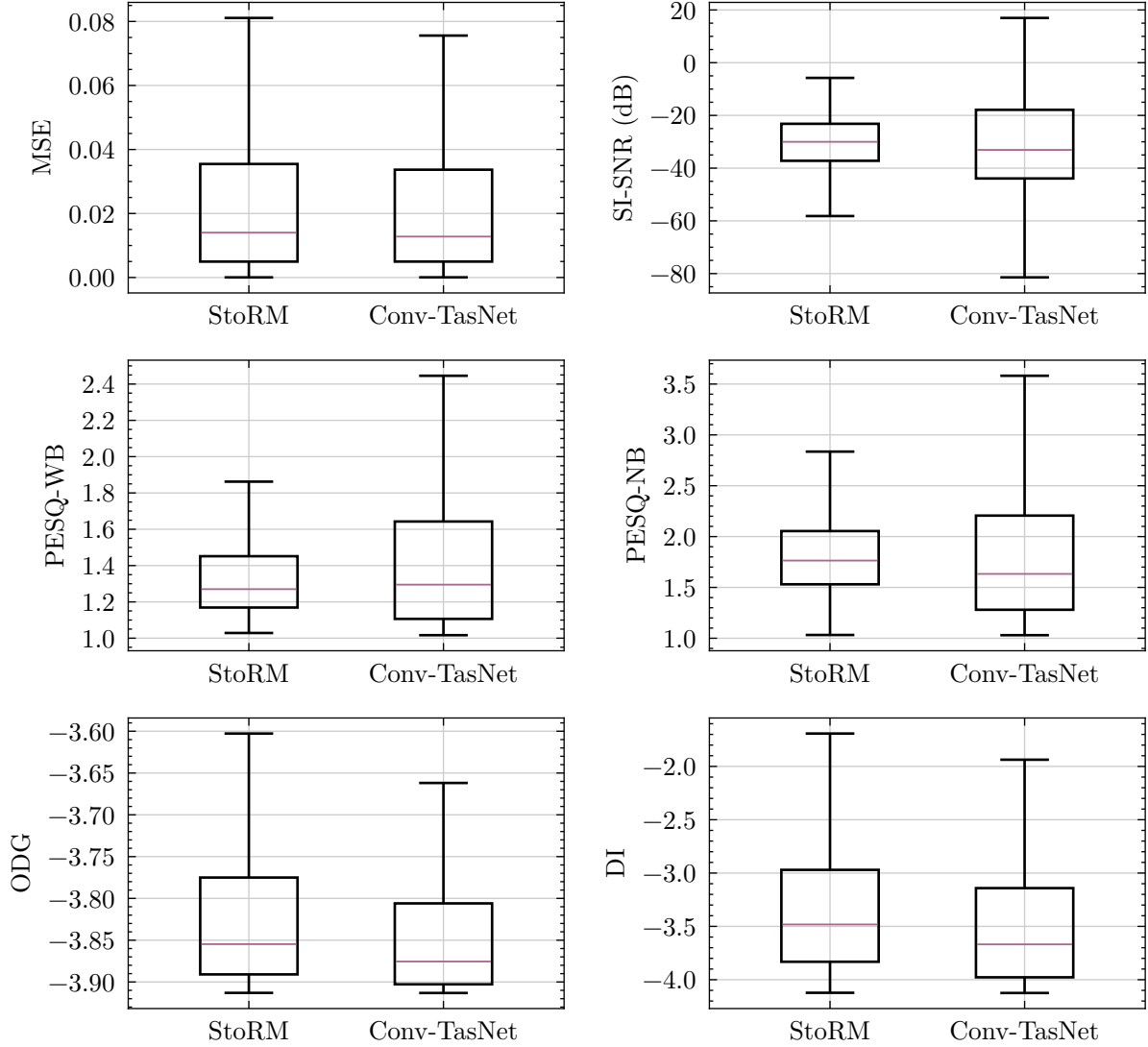


Figure 7.3: Boxplot comparison of different metrics for the evaluation of dereverberation performance of diverse audio samples. (Outliers not shown)

When comparing the two models against each other, the differences across most metrics are small relative to the standard deviation. Conv-TasNet achieves a marginally lower MSE (0.028 vs. 0.030) and slightly higher PESQ-WB (1.45 vs. 1.35), while StoRM scores better to a slight extent on ODG (-3.67 vs. -3.83) and DI (-3.14 vs. -3.50), suggesting it introduces fewer perceptual artifacts per sample on average. PESQ-NB is effectively equal (1.82 vs. 1.81). SI-SNR values are also close (-31.39 vs. -31.26), but with a notably higher standard deviation for both models, also indicating that performance is more inconsistent across samples.

The most unambiguous differentiator is computational cost: at comparable out-of-domain performance, Conv-TasNet processes all 2048 samples in 4 m 15 s (0.12 s per sample), while StoRM requires 6 h 14 m 51 s (10.98 s per sample) — approximately $91 \times$ the inference time. The samples were each 10 s long, meaning that Conv-TasNet processes audio roughly $83 \times$ faster than real time, while StoRM operates at approximately $0.91 \times$ real time, just below the threshold for real-time applicability, and notably on comparatively powerful hardware.

Overall, the differences between the two models are minimal and likely not perceptually significant, with both struggling to generalize to non-speech content.

7.7 Dereverberation Network

As mentioned in Section 5.4, a first implementation of the dereverberation network followed a purely generative approach, without masking operations. To validate the capacity of the model, a subset of the dataset was used to try and overfit this first implementation. This was successfully done, as seen in Figure 7.4, which shows a steady decrease in training and validation loss. The training loss reaches a value of 0.00005 and the validation loss of 0.0005, which is a strong indication that capacity is sufficient to learn the dereverberation of the training data. However, we could not replicate this behavior on the full dataset, meaning the model could not generalize well, which is why we switched to the masking approach described in Section 5.4.

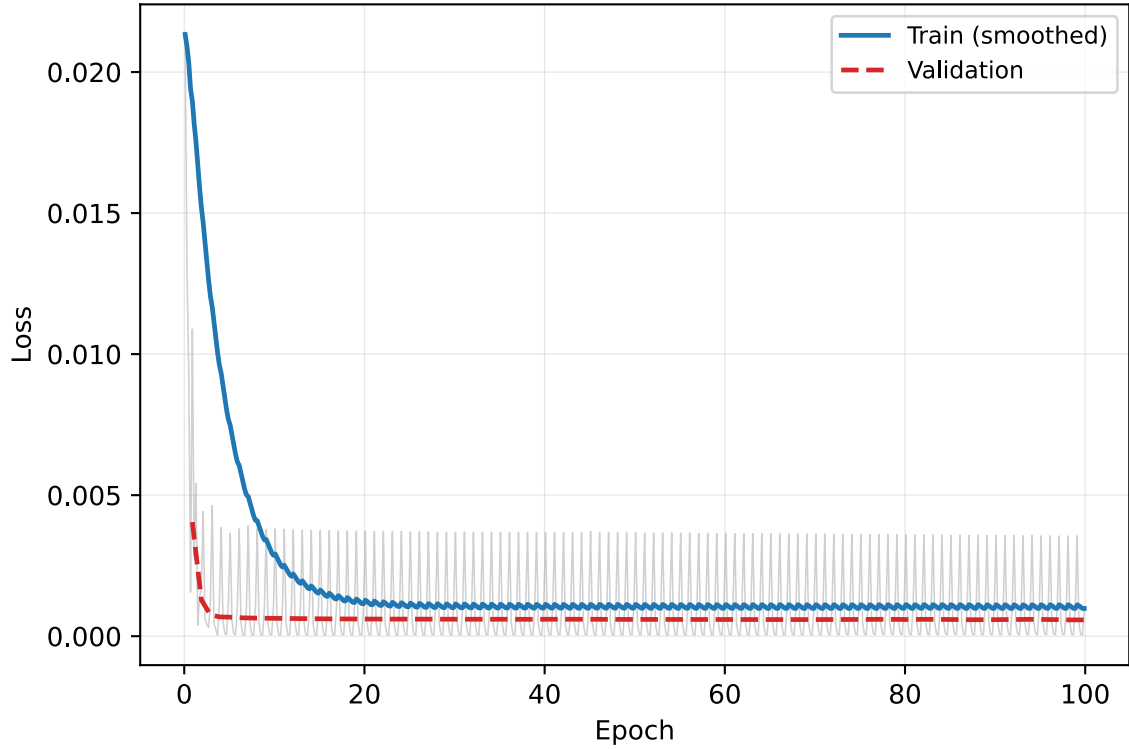


Figure 7.4: Loss curves of the first dereverberation network implementation, trained on a subset of the dataset using the perceptual quality network as a loss function.

Evaluation of the dereverberation network using the objective quality network was carried out sparsely, as the model’s output was not producing meaningful results. This can be seen in Figure 6.11, where the spectrum in the center shows the output, which barely resembles the clean reference on the right. The signal is distributed across the full model bandwidth (22.05 kHz), adding high-frequency noise, while the reverberation tail remains entirely unattenuated. The only discernible structure preserved from the clean signal is the gross syllabic rhythm; however, transient onsets are severely smeared across this otherwise incoherent spectrum. The precise cause of this phenomenon remains unclear. However, it can be hypothesized that, as discussed in Section 7.3.2, the shared training dataset between both models may be a contributing factor. Theoretically, a network exhibiting the performance characteristics shown in Section 6.4 should be sufficient, as comparable models have been successfully employed in prior work [Fu+21].

The dereverberation network trained with the SI-SNR loss was evaluated on a selected slice of the testing set using both spectrograms (Figure 6.9 and Figure 6.10) and objective metrics (Table 6.6), allowing for comparison with StoRM and Conv-TasNet.

In Figure 6.10, one can see a dereverberated speech signal in the center. Individual transients are clearly visible, and the rhythmic structure of the syllables is preserved. Reverberation tails are reduced in both magnitude and duration, though not fully eliminated. The spectral envelope remains close to that of the clean reference, indicating that the model does not introduce significant spectral distortion. An audible gating or pumping effect is noticeable, likely caused by the model suppressing regions between transients.

A further observation is that the model introduces additional high-frequency energy in some examples. This is consistent with the network having been trained on signals upsampled to 44.1 kHz, as described in Section 4.1.2.1. The model may have learned high-frequency patterns from the upsampling process rather than from genuine acoustic content. The resulting effect is occasionally perceived as a subtle enhancement, though it also introduces unwanted artifacts in other cases. This raises the question of whether reverberation should instead have been applied at the native 44.1 kHz prior to training.

One can see the effect of the additional 39 epochs of training when comparing Figure 6.9 and Figure 6.10. While the reverberation tail is only slightly further reduced in magnitude, the high-frequency noise is substantially reduced but still present. This is reflected in the SI-SNR improvement, which increases from 11.8 dB to 16.2 dB for the example shown.

The degree of dereverberation is closely tied to the characteristics of the input signal. Samples with little reverberation show only marginal improvement after processing, suggesting that the model scales its output to the perceived degree of reverberation in the input rather than applying a fixed transformation. More broadly, the perceptual quality of the enhanced signal is strongly dependent on

the quality of the input. Clean or mildly reverberant signals tend to be processed faithfully, while heavily degraded inputs yield less consistent results.

Table 6.6 shows a substantial SI-SNR improvement of +8.80 dB on average, which is slightly worse than the validation loss of 16.61 dB, but still confirms that the dereverberation network is able to generalize to unseen data. PESQ improves from 1.88 to 2.94, an increase of +1.06, indicating a meaningful gain in predicted speech quality. Notably, the standard deviation of both SI-SNR and PESQ is reduced in the enhanced signal relative to the baseline, suggesting that the model produces more consistent outputs across varying degrees of input reverberation.

The PEAQ metrics tell a different story. The ODG decreases marginally by -0.11 , and the DI shows a negligible change of $+0.01$, with both deltas exhibiting standard deviations larger than the mean shift itself. This is contrary to the improvements observed in SI-SNR and PESQ, again raising the question of whether PEAQ is a suitable metric for evaluating dereverberation performance, as discussed in Section 4.2. The MSE increase from 0.0010 to 0.0216 is a direct consequence of the SI-SNR training objective, which optimizes signal-to-noise ratio rather than sample-wise reconstruction fidelity, and is therefore expected.

Informal listening tests conducted on a selected set of samples yield the unexpected finding that music signals are well-suited to the proposed dereverberation approach. While the structural integrity of the recordings is preserved, reverberation-induced transient extensions and elongated decay times of percussive elements are effectively attenuated. Notably, this improvement remains perceptually noticeable even in fully mixed recordings, where the percussive components are masked by concurrent instrumental layers.

When compared against StoRM and Conv-TasNet, the dereverberation network achieves markedly superior performance on diverse audio signals (cf. Table 7.5), a result that is consistent with expectations given the domain-specific training. However, it remains an open question whether Conv-TasNet or StoRM would yield comparable performance if trained exclusively on the same dataset.

Table 7.5: Comparison of dereverberation performance of our model against StoRM and Conv-TasNet across in-domain speech and out-of-domain diverse audio samples.

Model	Speech	Diverse
Conv-TasNet	$15.3 \pm ?$ dB ³	-31.26 ± 18.6 dB
StoRM	6.5 ± 4.0 dB	-31.39 ± 11.71 dB
Ours	-	8.80 ± 3.02 dB

Evaluation of the dereverberation network led to the identification of multiple shortcomings. As discussed in Section 4.3, the motivation for developing the perceptual and objective quality networks extended beyond identifying an optimal indicator of dereverberation performance; it also eliminated the requirement for a clean reference signal during training. This improvement could have allowed us to train fully unsupervised on a larger subset of the AudioSet dataset, as it would have included many reverberant signals. This approach was not explored within the scope of this work.

The topic of reverberation being present in our dataset was touched upon in Section 7.3.2. It is discussed that the AudioSet and FSD50K datasets were not necessarily recorded in anechoic conditions. This can lead to mislabeling of data for use in training of the quality networks. Another problem arising is that the dereverberation network is shown echoic samples as ground truth, therefore not learning to eliminate reverberation but only reducing it. In the worst case, this occurs in 46.9% of instances (cf. Table 4.1). It can be hypothesized that a considerable portion of AudioSet samples originate from voiceover recordings, which are typically captured under near-anechoic conditions. Furthermore, FreeSound, as a sample-sharing platform, likely contains a high prevalence of acoustically dry recordings.

Further investigation is warranted to systematically compare datasets comprising exclusively anechoic recordings against those biased toward dry samples yet containing a non-negligible proportion of reverberant signals.

³SI-SNR taken from Luo and Mesgarani (2019), standard deviation not reported

8 Conclusion

This thesis set out to investigate state-of-the-art dereverberation models with respect to real-time applicability on diverse audio signals. Toward this goal, a systematic analysis of loss functions and their suitability as indicators of dereverberation performance was conducted, which motivated the development of a custom perceptual quality network for use as a training objective. A task-specific dataset was curated, ensuring a wide range of audio classes are covered, for which a diverse set of reverberation methods was implemented, namely RIR-based convolution, room simulation, and decay networks. This dataset demonstrated broad applicability, serving as an effective training resource for both the objective quality network and the dereverberation network. Building upon the Conv-TasNet architecture, a novel dereverberation model was subsequently implemented, incorporating both the SI-SNR and the proposed perceptual and objective quality network as loss functions, and evaluated against established baselines.

Our results show a gap in speech and out-of-domain dereverberation performance in the investigated existing models. StoRM, a frequency-domain model, and Conv-TasNet, a time-domain model, are both able to convincingly dereverberate speech signals but fail at out-of-domain samples such as music. Conv-TasNet demonstrated faster-than-real-time inference, rendering it markedly more computationally efficient than StoRM, for which only near-real-time performance could be achieved on prosumer-grade hardware.

The evaluation of loss functions investigated the MSE, MAE, SI-SNR, PESQ and PEAQ metrics as indicators of dereverberation. The SI-SNR is reported as the strongest metric. The perceptual quality network, primarily grounded in the ODG measure, failed to yield satisfactory performance. However, the objective quality network demonstrated good performance, surpassing the SI-SNR across the majority of evaluation conditions.

Our own implementation of a dereverberation network, utilizing SI-SNR, outperformed StoRM and Conv-TasNet on diverse signals. The same architecture used with the objective quality network was unable to generate usable results.

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A Appendix: Supplementary Material

A.1 Implementations of Dereverberation Network

Table A.1: All dereverberation hyperparameter configurations. Abbreviations: lr=learning rate, seg=segment length (seconds), N=encoder channels, k=kernel size, drop=dropout, X=num blocks per repeat, R=num repeats, dil=dilation pattern, sr=sample rate skip=skip connections enabled or not, loss=loss function used for training.

Run ID	lr	loss	seg	sr	N	win	tcn	k	drop	X	R	dil	causal	skip	Notes
derev_1	1e-3	perceptual	4.0	44100	64	-	[6x5]	4	0.1	0	0	-	true	false	Initial masking baseline
derev_1_increased	1e-4	perceptual	4.0	44100	265	-	-	3	0.0	8	3	-	false	true	Higher-capacity variant
derev_ten_v1	1e-4	perceptual	2.0	44100	256	-	[128x8]	3	0.1	8	4	1.128	false	true	First tuned TCN
derev_mha_v1	1e-4	perceptual	2.0	44100	-	-	-	-	-	-	-	-	-	-	Attention variant
derev_ten_v2_AdamW	1e-4	perceptual	2.0	44100	256	-	[128x8]	3	0.1	8	4	1.128	false	true	Optimizer study run
derev_ten_v2_MSE	1e-4	mse	2.0	44100	256	-	[128x8]	3	0.1	8	4	1.128	false	true	Loss ablation
derev_ten_v2_Small	1e-4	mse	2.0	44100	256	-	[128x6]	3	0.1	6	4	1.32	false	true	Smaller TCN
derev_ten_v2_Dropout	1e-4	perceptual	2.0	44100	256	-	[128x8]	3	0.2	8	4	1.128	false	true	Dropout ablation
derev_ten_v3	1e-4	perceptual	2.0	44100	256	-	[128x8]	3	0.1	8	4	1.128	false	true	v3 baseline
derev_ten_v3_MSE	1e-4	mse	2.0	44100	256	-	[128x8]	3	0.1	8	4	1.128	false	true	v3 with MSE
derev_convtaenet	1e-4	perceptual	2.0	44100	512	2ms	[512x24]	3	0.0	8	3	1.128x3	false	true	Conv-TasNet-style setting
derev_ten_v4_SISNR	1e-4	sisnr	4.0	44100	512	2ms	[179x24]	3	0.0	8	3	1.128x3	false	true	v4 with SI-SNR loss
derev_ten_v4_percep	1e-4	perceptual	4.0	44100	512	2ms	[179x24]	3	0.0	8	3	1.128x3	false	true	v4 with perceptual loss

A.2 Results of Perceptual Quality Network

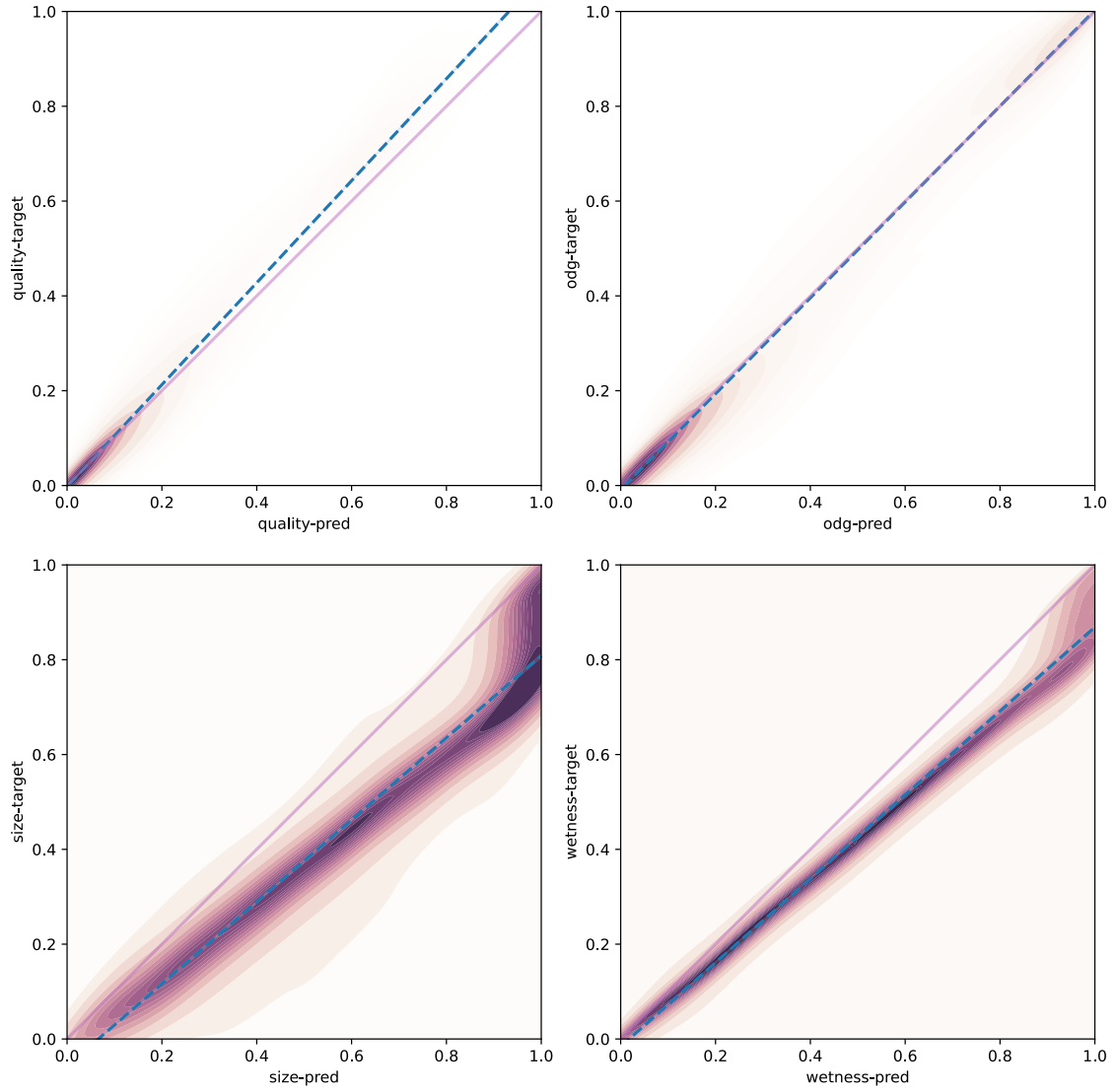


Figure A.1: KDE plots of predicted versus ground truth values of all perceptual quality metrics at epoch 195, with the diagonal indicating perfect agreement.

A.3 Results of Objective Quality Network

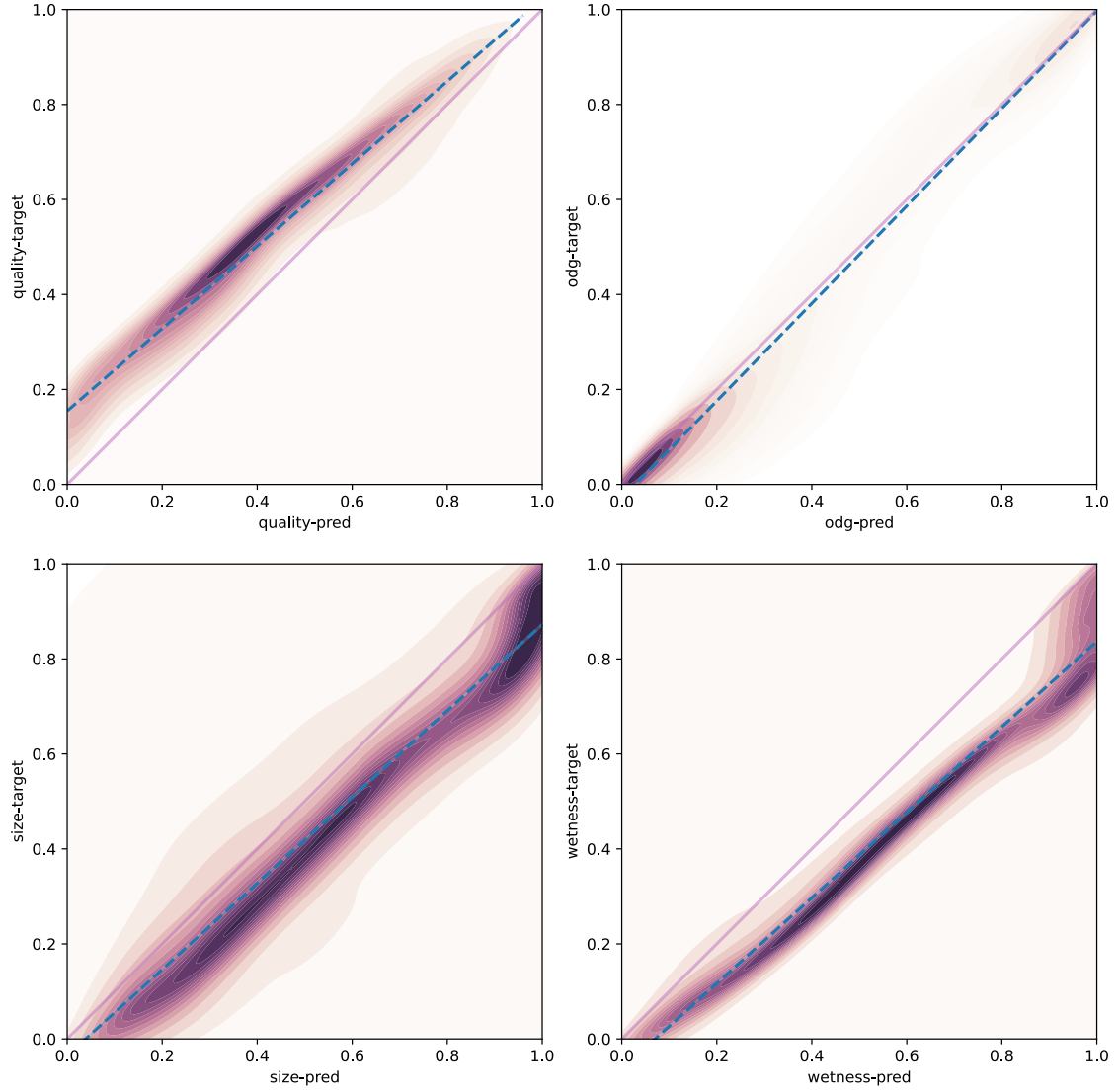


Figure A.2: KDE plots of predicted versus ground truth values of all objective quality metrics using the updated quality score at epoch 61, with the diagonal indicating perfect agreement.

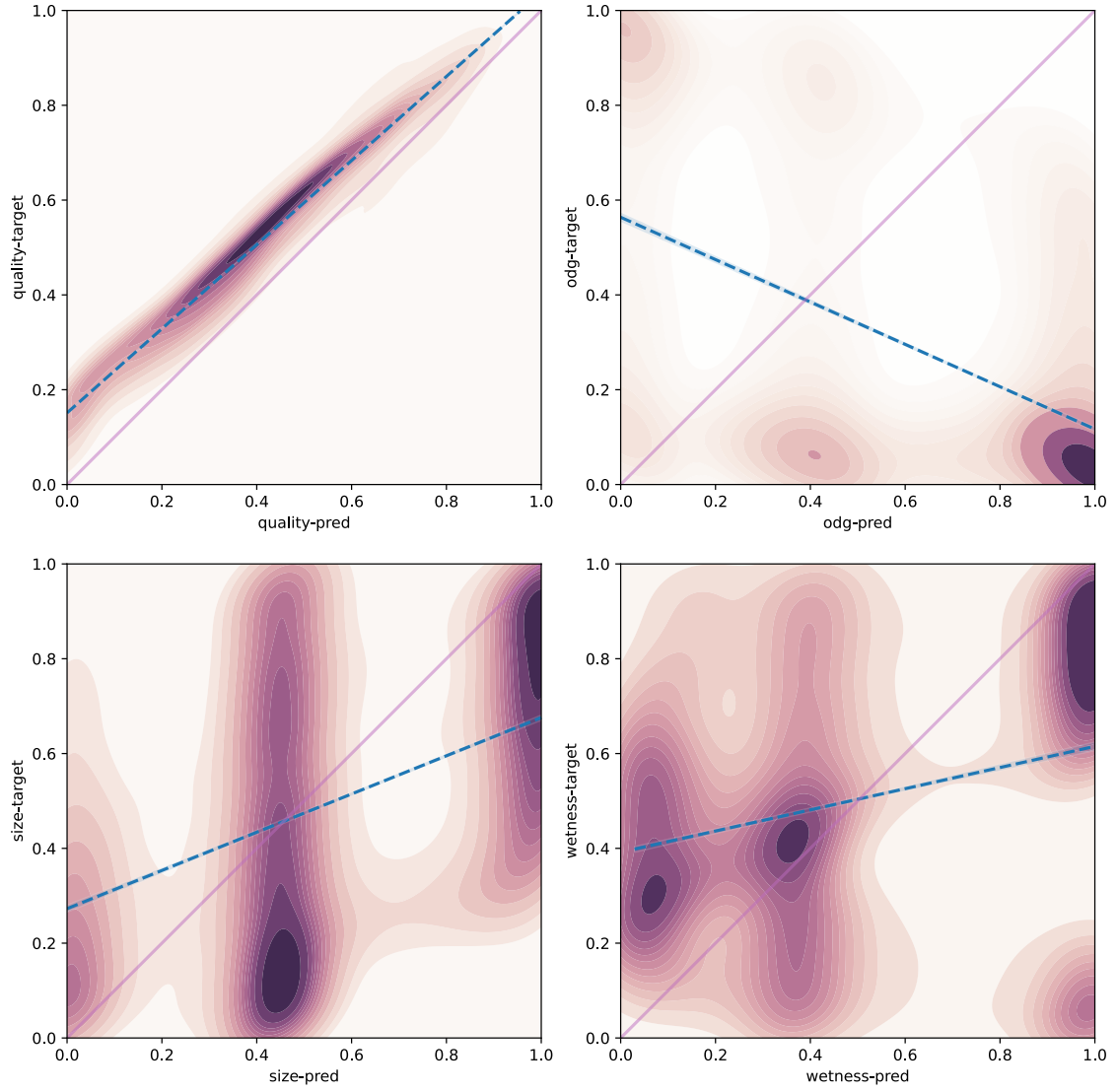


Figure A.3: KDE plots of predicted versus ground truth values of all objective quality metrics using the updated quality score and loss at epoch 166, with the diagonal indicating perfect agreement.